CaSB 150 Midterm: Practice Problem Bank

Fully-Detailed Worked Solutions — Every Step Shown

Lectures 1–9 • Based on 2017–2024 Past Exams

Algebraic operations are labeled in blue italics so you can see exactly which manipulation was applied at each step.

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1. Growth Equations & Taylor Series

1.1 Problem 1.1: Modified Exponential Growth (2023-style)

Problem

Consider growth governed by

$$\frac{dN(t)}{dt} = rN(t)\left(1 - e^{-N(t)/N(\tau)}\right)$$

where τ is a time constant and $N(\tau)$ is a reference size.

(a) At early times when $N(t) \ll N(\tau)$, simplify and solve.

(b) At long times when $N(t) \gg N(\tau)$, simplify and solve.

(c) What are the units of r? Is it a per-capita growth rate?

(d) Sketch dN/dt vs t.

(e) Find the linear space (if it exists) for part (a).

(f) Does this equation have a maximum population K?

Solution

(a) Early times, $N \ll N(\tau)$:
Step 1. Identify the small quantity. We have $x \equiv N/N(\tau) \ll 1$.
Step 2. Apply the Taylor expansion of e^{-x} around $x = 0$:

$$e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \cdots$$

(Taylor series of e^{-x})
Step 3. Substitute $x = N/N(\tau)$:

$$e^{-N/N(\tau)} = 1 - \frac{N}{N(\tau)} + \frac{1}{2}\left(\frac{N}{N(\tau)}\right)^2 - \cdots$$

Step 4. Compute $1 - e^{-N/N(\tau)}$ by distributing the negative sign:

$$1 - e^{-N/N(\tau)} = 1 - 1 + \frac{N}{N(\tau)} - \frac{1}{2}\left(\frac{N}{N(\tau)}\right)^2 + \cdots$$

($1 - 1 = 0$, sign of each term flips)
Step 5. Keep only the leading (lowest-order) term:

$$1 - e^{-N/N(\tau)} \approx \frac{N}{N(\tau)}$$

Step 6. Substitute back into the ODE:

$$\frac{dN}{dt} \approx rN \cdot \frac{N}{N(\tau)} = \frac{r}{N(\tau)}N^2$$

This is super-exponential (power-law) growth with $p = 2$.
Step 7. Separate variables:

$$\frac{dN}{N^2} = \frac{r}{N(\tau)}\,dt$$

(divide both sides by N^2, multiply both sides by dt)
Step 8. Integrate both sides:

$$\int N^{-2}\,dN = \int \frac{r}{N(\tau)}\,dt$$

Step 9. Apply the power rule $\int N^{-2}dN = N^{-1}/(-1) = -1/N$:

$$-\frac{1}{N} = \frac{rt}{N(\tau)} + C$$

Step 10. Apply IC $N(0) = N_0$ at $t = 0$:

$$-\frac{1}{N_0} = 0 + C \quad \Longrightarrow \quad C = -\frac{1}{N_0}$$

Step 11. Substitute C back and multiply both sides by -1:

$$\frac{1}{N} = \frac{1}{N_0} - \frac{rt}{N(\tau)}$$

Step 12. Get common denominator on the right:

$$\frac{1}{N} = \frac{N(\tau) - rN_0 t}{N_0 N(\tau)}$$

(multiply $1/N_0$ by $N(\tau)/N(\tau)$ and $rt/N(\tau)$ by N_0/N_0)
Step 13. Take reciprocal:

$$\boxed{N(t) = \frac{N_0\, N(\tau)}{N(\tau) - rN_0 t}}$$

Step 14. Blow-up time: denominator $= 0$:

$$N(\tau) - rN_0 t^* = 0 \quad \Longrightarrow \quad t^* = \frac{N(\tau)}{rN_0}$$

(b) Long times, $N \gg N(\tau)$:
Step 1. When $N/N(\tau)$ is large, $e^{-N/N(\tau)} \to 0$ (exponential decay):

$$1 - e^{-N/N(\tau)} \to 1 - 0 = 1$$

Step 2. The ODE becomes:

$$\frac{dN}{dt} \approx rN \cdot 1 = rN$$

Step 3. Separate variables and integrate:

$$\frac{dN}{N} = r\,dt \quad \Longrightarrow \quad \int \frac{dN}{N} = \int r\,dt \quad \Longrightarrow \quad \ln N = rt + C'$$

Step 4. Exponentiate both sides:

$$N = e^{rt+C'} = e^{C'} \cdot e^{rt}$$

(using $e^{a+b} = e^a e^b$)
Step 5. Rename $e^{C'} \to N_0^*$:

$$\boxed{N(t) \approx N_0^* e^{rt}}$$

Standard exponential growth.

(c) Units of r:
Step 1. The factor $(1 - e^{-N/N(\tau)})$ is dimensionless (exponential of a dimensionless ratio).
Step 2. LHS dN/dt has units of $[N]/[t]$. So rN must have units of $[N]/[t]$, giving:

$$[r] = \frac{[N]/[t]}{[N]} = \frac{1}{[t]} = \text{time}^{-1}$$

Step 3. **Yes**, r is a per-capita rate (confirmed by long-time limit $\dot{N} = rN$).

(d) Sketch: Early times: $dN/dt \sim rN^2/N(\tau)$ blows up like $1/(t^* - t)^2$. Long times: $dN/dt \sim rN_0^* e^{rt}$ ordinary exponential.

(e) Linear space for part (a):
Step 1. Try $u = 1/N$. Compute du/dt using chain rule:

$$\frac{du}{dt} = \frac{d(N^{-1})}{dt} = -N^{-2}\frac{dN}{dt}$$

(chain rule: $\frac{d}{dt}[N^{-1}] = -N^{-2} \cdot \dot{N}$)
Step 2. Substitute $\dot{N} = (r/N(\tau))N^2$:

$$\frac{du}{dt} = -N^{-2} \cdot \frac{r}{N(\tau)} \cdot N^2 = -\frac{r}{N(\tau)}$$

($N^{-2} \cdot N^2 = 1$)
Step 3. The result is constant! So $1/N$ vs t is the linear space.

$$\boxed{\text{Linear space: } 1/N \text{ vs. } t}$$

(f) Maximum K:
This equation has no upper bound (the long-time limit blows up exponentially). To add one, multiply by a logistic factor:

$$\frac{dN}{dt} = rN(1 - e^{-N/N(\tau)})\left(1 - \frac{N}{K}\right)$$

1.2 Problem 1.2: Power-of-Time Growth (2019/2021-style)

Problem

Consider $\frac{dN(t)}{dt} = r\frac{N(t)}{t}$.

(a) Solve for $N(t)$.

(b) Interpret r and find its units.

(c) Will growth ever stop?

(d) For $r = 10$, compute the 4th-order Taylor term about $t_0 = 1$.

(e) Find a mathematical space where Euler's method is exact.

Solution

(a) Solve for $N(t)$:
Step 1. Separate variables:

$$\frac{dN}{N} = \frac{r\,dt}{t}$$

(divide both sides by N, multiply by dt)
Step 2. Integrate both sides:

$$\int \frac{dN}{N} = r\int \frac{dt}{t}$$

(r is constant, factor out)
Step 3. Apply $\int dx/x = \ln x$:

$$\ln N = r\ln t + C$$

Step 4. Apply log power rule $r\ln t = \ln(t^r)$:

$$\ln N = \ln(t^r) + C$$

Step 5. Exponentiate both sides:

$$e^{\ln N} = e^{\ln(t^r)+C} = e^{\ln(t^r)} \cdot e^C$$

(splitting $e^{a+b} = e^a e^b$)

Step 6. Simplify using $e^{\ln x} = x$:

$$N = t^r \cdot e^C$$

Step 7. Apply IC at $t = 1$: $N(1) = 1^r \cdot e^C = 1 \cdot e^C = e^C$, so $e^C = N(1)$:

$$\boxed{N(t) = N(1)\, t^r}$$

(b) Interpret r:

Step 1. Check units in original ODE: dN/dt has units $[N]/[t]$. RHS is rN/t, with N/t having units $[N]/[t]$. So r is dimensionless.

Step 2. **r is dimensionless — an elasticity, not a rate.** If t doubles, N scales by 2^r. Equal proportional changes in t produce equal proportional changes in N.

(c) Will growth stop?

Step 1. Set $\dot{N} = 0$: $rN/t = 0$ requires $N = 0$ or $t = \infty$.

Step 2. For $r > 0$ and finite t, $N(t) = N(1)t^r \to \infty$ as $t \to \infty$. **No finite-time fixed point; growth never stops.**

(d) 4th-order Taylor term for $r = 10$:

Step 1. First derivative using power rule $\frac{d}{dt}[t^r] = rt^{r-1}$:

$$N'(t) = N(1) \cdot r \cdot t^{r-1} = 10\, N(1)\, t^9$$

($N(1)$ is constant; $r = 10$ gives exponent $r - 1 = 9$)

Step 2. Second derivative:

$$N''(t) = 10 \cdot 9\, N(1)\, t^8 = 90\, N(1)\, t^8$$

Step 3. Third derivative:

$$N'''(t) = 90 \cdot 8\, N(1)\, t^7 = 720\, N(1)\, t^7$$

Step 4. Fourth derivative:

$$N^{(4)}(t) = 720 \cdot 7\, N(1)\, t^6 = 5040\, N(1)\, t^6$$

Step 5. Evaluate at $t_0 = 1$:

$$N^{(4)}(1) = 5040\, N(1) \cdot 1^6 = 5040\, N(1)$$

Step 6. The 4th-order Taylor term is $N^{(4)}(t_0)(\Delta t)^4/4!$:

$$\frac{N^{(4)}(1)}{4!}(\Delta t)^4 = \frac{5040\, N(1)}{24}(\Delta t)^4$$

Step 7. Compute $5040/24$: $5040/24 = 210$.

$$\boxed{\text{4th-order term} = 210\, N(1)\, (\Delta t)^4}$$

All four derivatives exist and are non-zero, so RK4 improves over Euler.

(e) Linear space:

Step 1. Try $u = \ln N$, $s = \ln t$. Compute du/ds via the chain rule trick:

$$\frac{du}{ds} = \frac{d \ln N}{d \ln t} = \frac{d \ln N/dt}{d \ln t/dt}$$

Step 2. Compute $d \ln N/dt$:

$$\frac{d \ln N}{dt} = \frac{1}{N}\frac{dN}{dt} = \frac{1}{N} \cdot \frac{rN}{t} = \frac{r}{t}$$

(chain rule on $\ln N$, then substitute $\dot{N} = rN/t$, N's cancel)

Step 3. Compute $d\ln t/dt$:

$$\frac{d\ln t}{dt} = \frac{1}{t}$$

Step 4. Divide:

$$\frac{du}{ds} = \frac{r/t}{1/t} = \frac{r}{t} \cdot \frac{t}{1} = r$$

(dividing by $1/t$ is multiplying by t, t's cancel)

Step 5. The result is constant.

$$\boxed{\text{Linear space: } \ln N \text{ vs. } \ln t \text{ (log-log)}}$$

Key Insight

Finding the "perfect space": ask what variable transformation makes $d(\text{thing})/d(\text{other thing}) = \text{constant}$. Common transformations:

- Exponential growth $\dot{N} = rN$: $\ln N$ vs t (semi-log)
- Power-law in time $\dot{N} = rN/t$: $\ln N$ vs $\ln t$ (log-log)
- Inverse-time $\dot{N} = r/t$: N vs $\ln t$ (semi-log on t)
- Power $\dot{N} = rN^2$: $1/N$ vs t

1.3 Problem 1.3: Discrete vs. Continuous Conversion

Problem

A bacterial colony starts at $N(0) = 100$. Each generation produces an average of $r_d = 4$ offspring per parent (so the next generation is $5\times$ the current). Generation time is 30 minutes.

(a) Write discrete and continuous growth equations.

(b) Find the continuous per-capita rate r_c.

(c) How many bacteria after 5 hours, using each method?

(d) For mixed-rate growth: $r_1 = 0.05/\text{min}$ for first 60 min, $r_2 = 0.15/\text{min}$ for next 60 min, find N at $t = 120$ min and effective rate.

Solution

(a) Equations:

Discrete: Each generation, population multiplies by $1 + r_d = 1 + 4 = 5$:

$$N(t+1) = (1 + r_d)N(t) = 5N(t)$$

Solution: $N(g) = 100 \cdot 5^g$ where g counts generations.

Continuous: per-capita rate r_c:

$$\frac{dN}{dt} = r_c N$$

Solution: $N(t) = 100\, e^{r_c t}$.

(b) Find r_c:

Step 1. Continuous and discrete must agree over one generation. After $\tau_{\text{gen}} = 30$ min, both should give the same multiplier $\lambda = 5$:

$$e^{r_c \cdot \tau_{\text{gen}}} = \lambda$$

Step 2. Substitute values:

$$e^{r_c \cdot 30} = 5$$

Step 3. Take natural log of both sides:

$$\ln(e^{30r_c}) = \ln 5$$

($\ln(e^x) = x$)

$$30r_c = \ln 5$$

Step 4. Solve for r_c:

$$r_c = \frac{\ln 5}{30}$$

Step 5. Numerically: $\ln 5 \approx 1.609$, so:

$$r_c \approx \frac{1.609}{30} \approx 0.0536\,\text{min}^{-1}$$

(c) After 5 hours:
Step 1. Convert: 5 hr $= 5 \times 60 = 300$ min.
Step 2. Number of generations: $300/30 = 10$ generations.
Step 3. Discrete: $N(10) = 100 \cdot 5^{10}$.
Step 4. Compute 5^{10}:

$$5^2 = 25, \quad 5^4 = 25^2 = 625, \quad 5^5 = 5 \cdot 5^4 = 3125$$

$$5^{10} = 5^5 \cdot 5^5 = 3125 \cdot 3125 = 9{,}765{,}625$$

Step 5. Multiply by 100:

$$N = 100 \cdot 9{,}765{,}625 = 9.77 \times 10^8$$

Step 6. Continuous check: $N(300) = 100\,e^{0.0536 \cdot 300} = 100\,e^{16.09}$.

$$e^{16.09} \approx 9.77 \times 10^6 \quad \Longrightarrow \quad N(300) \approx 9.77 \times 10^8 \checkmark$$

Same answer.

(d) Mixed-rate:
Step 1. First 60 min at rate $r_1 = 0.05$:

$$N(60) = 100\,e^{r_1 \cdot 60} = 100\,e^{0.05 \cdot 60} = 100\,e^3$$

Step 2. Compute $e^3 \approx 20.09$:

$$N(60) \approx 100 \cdot 20.09 = 2009$$

Step 3. Next 60 min at rate $r_2 = 0.15$:

$$N(120) = N(60) \cdot e^{r_2 \cdot 60} = 2009 \cdot e^{0.15 \cdot 60} = 2009 \cdot e^9$$

Step 4. Compute $e^9 \approx 8103$:

$$N(120) \approx 2009 \cdot 8103 \approx 1.63 \times 10^7$$

Step 5. Effective rate is the time-weighted average:

$$r_{\text{eff}} = \frac{r_1 t_1 + r_2 t_2}{t_1 + t_2} = \frac{0.05 \cdot 60 + 0.15 \cdot 60}{120} = \frac{3+9}{120} = \frac{12}{120} = 0.10\,\text{min}^{-1}$$

Step 6. Verify: $100\,e^{0.10 \cdot 120} = 100\,e^{12}$. Compute $e^{12} \approx 162754$:

$$100 \cdot 162754 \approx 1.63 \times 10^7 \checkmark$$

1.4 Problem 1.4: Super-Exponential Growth (2018-style)

Problem

$\frac{dN}{dt} = rN^3$ with $N(0) = 2$, $r = 0.01$.

(a) Write the discrete version.

(b) Solve the continuous equation.

(c) At what time does N go to infinity?

Solution

(a) Discrete version (Euler with $\Delta t = 1$):
Step 1. The Euler approximation is $N(t + \Delta t) \approx N(t) + \dot{N} \cdot \Delta t$.
Step 2. Substitute $\dot{N} = rN^3$ and $\Delta t = 1$:

$$N(t+1) = N(t) + rN(t)^3$$

(b) Solve continuous equation:
Step 1. Separate variables:

$$\frac{dN}{N^3} = r\,dt$$

(divide both sides by N^3, multiply by dt)
Step 2. Rewrite $1/N^3 = N^{-3}$:

$$N^{-3}\,dN = r\,dt$$

Step 3. Integrate both sides:

$$\int N^{-3}\,dN = \int r\,dt$$

Step 4. Apply power rule: $\int N^{-3}dN = N^{-2}/(-2) = -1/(2N^2)$:

$$-\frac{1}{2N^2} = rt + C$$

Step 5. Apply IC $N(0) = N_0$ at $t = 0$:

$$-\frac{1}{2N_0^2} = 0 + C \quad \Longrightarrow \quad C = -\frac{1}{2N_0^2}$$

Step 6. Substitute back:

$$-\frac{1}{2N^2} = rt - \frac{1}{2N_0^2}$$

Step 7. Multiply both sides by -2:

$$\frac{1}{N^2} = \frac{1}{N_0^2} - 2rt$$

Step 8. Common denominator on RHS:

$$\frac{1}{N^2} = \frac{1 - 2rN_0^2 t}{N_0^2}$$

Step 9. Take reciprocal:

$$N^2 = \frac{N_0^2}{1 - 2rN_0^2 t}$$

Step 10. Take square root:

$$\boxed{N(t) = \frac{N_0}{\sqrt{1 - 2rN_0^2\,t}}}$$

(c) Blow-up time:

Step 1. $N \to \infty$ when denominator $\to 0$:

$$1 - 2rN_0^2 t^* = 0$$

Step 2. Solve for t^*:

$$t^* = \frac{1}{2rN_0^2}$$

Step 3. Plug in $N_0 = 2$, $r = 0.01$:

$$t^* = \frac{1}{2(0.01)(2)^2} = \frac{1}{2(0.01)(4)} = \frac{1}{0.08} = 12.5$$

1.5 Problem 1.5: Logistic Growth Full Treatment

Problem

$\frac{dN}{dt} = rN(1 - N/K)$.

(a) Find fixed points and check stability.

(b) Solve exactly.

(c) Sketch $N(t)$ vs t for $N_0 < K/2$, $N_0 = K/2$, $N_0 > K$.

Solution

(a) Fixed points and stability:

Step 1. Set $\dot{N} = 0$:

$$rN(1 - N/K) = 0$$

Step 2. Apply zero-product property:

$$N = 0 \quad \text{OR} \quad 1 - N/K = 0$$

Step 3. Solve the second case:

$$1 - N/K = 0 \quad \Longrightarrow \quad N/K = 1 \quad \Longrightarrow \quad N = K$$

So fixed points: $N^* = 0$ and $N^* = K$.

Step 4. Linearize. Define $f(N) = rN(1 - N/K) = rN - rN^2/K$. Differentiate:

$$f'(N) = r - \frac{2rN}{K}$$

$(\frac{d}{dN}[rN^2/K] = 2rN/K)$

Step 5. Evaluate at $N^* = 0$:

$$f'(0) = r - 0 = r > 0 \quad \Longrightarrow \quad \textbf{unstable}$$

Step 6. Evaluate at $N^* = K$:

$$f'(K) = r - \frac{2rK}{K} = r - 2r = -r < 0 \quad \Longrightarrow \quad \textbf{stable}$$

(b) Solve exactly via partial fractions:

Step 1. Separate variables:

$$\frac{dN}{N(1 - N/K)} = r\,dt$$

Step 2. Decompose the LHS via partial fractions. Write:

$$\frac{1}{N(1 - N/K)} = \frac{A}{N} + \frac{B}{1 - N/K}$$

Step 3. Multiply both sides by $N(1 - N/K)$:

$$1 = A(1 - N/K) + BN$$

Step 4. Set $N = 0$:

$$1 = A(1 - 0) + 0 \quad \Longrightarrow \quad A = 1$$

Step 5. Set $N = K$:

$$1 = A \cdot 0 + BK \quad \Longrightarrow \quad B = 1/K$$

Step 6. So:

$$\frac{1}{N(1 - N/K)} = \frac{1}{N} + \frac{1/K}{1 - N/K}$$

Step 7. Integrate both sides:

$$\int \left(\frac{1}{N} + \frac{1/K}{1 - N/K} \right) dN = \int r\, dt$$

Step 8. Integrate the first term: $\int dN/N = \ln N$.
Step 9. For the second term, substitute $u = 1 - N/K$, $du = -dN/K$, so $dN = -K\, du$:

$$\int \frac{1/K}{u} \cdot (-K\, du) = -\int \frac{1}{u}\, du = -\ln|u| = -\ln(1 - N/K)$$

($1/K \cdot (-K) = -1$, then standard $\int du/u = \ln u$)
Step 10. Combined:

$$\ln N - \ln(1 - N/K) = rt + C$$

Step 11. Apply log subtraction rule $\ln A - \ln B = \ln(A/B)$:

$$\ln \frac{N}{1 - N/K} = rt + C$$

Step 12. Exponentiate both sides:

$$\frac{N}{1 - N/K} = e^{rt+C} = e^C \cdot e^{rt}$$

($e^{a+b} = e^a e^b$)
Step 13. Apply IC at $t = 0$:

$$\frac{N_0}{1 - N_0/K} = e^C \quad \Longrightarrow \quad e^C = \frac{N_0 K}{K - N_0}$$

(multiply numerator and denominator by K to clear the fraction in denominator)
Step 14. Substitute back:

$$\frac{N}{1 - N/K} = \frac{N_0 K}{K - N_0} \cdot e^{rt}$$

Step 15. Solve for N. Let $A = \frac{N_0 K}{K - N_0} \cdot e^{rt}$. Then $N = A(1 - N/K)$:

$$N = A - \frac{A}{K} N$$

Step 16. Move all N terms to one side:

$$N + \frac{A}{K} N = A \quad \Longrightarrow \quad N \left(1 + \frac{A}{K} \right) = A$$

Step 17. Solve for N:

$$N = \frac{A}{1 + A/K} = \frac{AK}{K + A}$$

(multiply numerator and denominator by K)

Step 18. Substitute A back and simplify. After algebra (multiplying by $(K - N_0)/(K - N_0)e^{-rt}$ etc.):

$$\boxed{N(t) = \frac{K}{1 + \left(\frac{K - N_0}{N_0}\right) e^{-rt}}}$$

Step 19. Sanity check: as $t \to \infty$, $e^{-rt} \to 0$, so $N \to K/(1+0) = K$. ✓

(c) Sketch interpretation:

- $N_0 < K/2$: S-shape (sigmoid). Slow at first, fast through middle, slow approaching K.
- $N_0 = K/2$: starts at the inflection point; concave-down approach.
- $N_0 > K$: exponentially decaying approach to K from above.

The inflection point at $N = K/2$ is where $\dot{N}$ is maximized (max growth rate).

1.6 Problem 1.6: Inverse-Time Growth (2021 Q1)

Problem

Consider $\dfrac{dN(t)}{dt} = \dfrac{r}{t}$.

(a) Solve for $N(t)$.

(b) What are the units of r? Is r a per-capita growth rate?

(c) When $r > 0$, does the population ever reach a stable finite size?

(d) For $r = 10$, compute the 4th-order Taylor term about $t_0 = 1$.

(e) Find the linear space (perfect Euler).

(f) Compare to power-of-time growth $\dot{N} = rN/t$: what is the key difference?

Solution

(a) Solve:
Step 1. The equation is already separated ($\dot{N}$ depends only on t, not N):

$$dN = \frac{r}{t}\,dt$$

Step 2. Integrate both sides directly:

$$\int dN = r \int \frac{dt}{t} \quad \Longrightarrow \quad N = r \ln t + C$$

Step 3. Apply IC at $t = 1$: $N(1) = r \ln 1 + C = 0 + C = C$, so $C = N(1)$:

$$\boxed{N(t) = N(1) + r \ln t}$$

(b) Units of r:
Step 1. Check dimensions. LHS dN/dt has units [population]/[time]. RHS r/t has units $[r]$/[time].
Step 2. Matching: $[r]$/[time] = [population]/[time], so $[r]$ = population.
Step 3. r has **units of population, not time^{-1}**. It is **not** a per-capita growth rate (per-capita rates have units of 1/time).

(c) Stable finite size?
Step 1. Set $\dot{N} = 0$: $r/t = 0$ requires $t \to \infty$ (for $r \neq 0$).
Step 2. For $r > 0$: $N(t) = N(1) + r \ln t \to \infty$ as $t \to \infty$. **Growth never stops.**

Step 3. However, growth is very slow (logarithmic). Compare: exponential $N \sim e^{rt}$; power $N \sim t^r$; inverse-time $N \sim r \ln t$. Inverse-time is the slowest growing of all.

(d) 4th-order Taylor term for $r = 10$:
Step 1. $N(t) = N(1) + 10 \ln t$. Compute derivatives (using $\frac{d}{dt}[\ln t] = t^{-1}$, $\frac{d}{dt}[t^{-1}] = -t^{-2}$, etc.):

$$N'(t) = \frac{10}{t} = 10t^{-1}$$

$$N''(t) = -10t^{-2}$$

$$N'''(t) = 20t^{-3}$$

$$N^{(4)}(t) = -60t^{-4}$$

Step 2. Evaluate at $t_0 = 1$: $N^{(4)}(1) = -60$.
Step 3. 4th-order term:

$$\frac{N^{(4)}(1)}{4!}(\Delta t)^4 = \frac{-60}{24}(\Delta t)^4 = -\frac{5}{2}(\Delta t)^4$$

All four derivatives exist, so RK4 improves over Euler.

(e) Linear space:
Step 1. Try $u = N$, $s = \ln t$. Compute du/ds:

$$\frac{du}{ds} = \frac{dN}{d\ln t} = \frac{dN/dt}{d\ln t/dt} = \frac{r/t}{1/t} = r$$

($d\ln t/dt = 1/t$; dividing r/t by $1/t$ gives r)
Step 2. Constant! So N vs $\ln t$ is the linear space (N **vs semi-log** t).

(f) Comparison with power-of-time:

Property	Power-of-time $\dot{N} = rN/t$	Inverse-time $\dot{N} = r/t$
Solution	$N(t) = N(1)t^r$	$N(t) = N(1) + r\ln t$
Units of r	Dimensionless	Population
Linear space	$\ln N$ vs $\ln t$	N vs $\ln t$
Depends on N?	Yes ($\dot{N} \propto N$)	No ($\dot{N}$ independent of N)
Growth for $r > 0$	Power law (faster)	Logarithmic (slower)

Key difference: Power-of-time is a per-capita equation (growth proportional to N); inverse-time is an absolute rate (same increase regardless of how many organisms exist).

1.7 Problem 1.7: Piecewise Growth Rates (2017 Q1)

Problem

A population grows at rate r_1 for time t_1, then immediately shifts to rate r_2 for time t_2. The initial population is $N(0)$.

(a) Write $N(t_1 + t_2)$ as a single exponential.

(b) If $r_1 = 0.1$/month, $r_2 = 0.2$/month, $t_1 = 10$ months, $t_2 = 26$ months, $N(0) = 2$, what is $N(t_1 + t_2)$?

(c) Define and compute the effective growth rate r_{eff}.

(d) A discrete population has $r_d = 6$ offspring per parent, with generation time 1 year. Find the equivalent continuous r_c.

Solution

(a) Single exponential expression:

Step 1. During phase 1, standard exponential growth:

$$N(t_1) = N(0)\, e^{r_1 t_1}$$

Step 2. This $N(t_1)$ is the initial condition for phase 2:

$$N(t_1 + t_2) = N(t_1)\, e^{r_2 t_2} = N(0)\, e^{r_1 t_1} \cdot e^{r_2 t_2}$$

Step 3. Combine using $e^a \cdot e^b = e^{a+b}$:

$$\boxed{N(t_1 + t_2) = N(0)\, e^{r_1 t_1 + r_2 t_2}}$$

(b) Numerical evaluation:
Step 1. Compute exponent: $r_1 t_1 + r_2 t_2 = 0.1(10) + 0.2(26) = 1 + 5.2 = 6.2$.
Step 2.

$$N(36 \text{ months}) = 2\, e^{6.2}$$

Step 3. Compute $e^{6.2} \approx 492.7$:

$$N \approx 2 \times 492.7 \approx 985 \text{ individuals}$$

(c) Effective growth rate:
Step 1. The effective rate satisfies $N(0)e^{r_{\text{eff}}(t_1+t_2)} = N(0)e^{r_1 t_1 + r_2 t_2}$.
Step 2. Equate exponents:

$$r_{\text{eff}}(t_1 + t_2) = r_1 t_1 + r_2 t_2$$

Step 3. Divide both sides by $(t_1 + t_2)$:

$$\boxed{r_{\text{eff}} = \frac{r_1 t_1 + r_2 t_2}{t_1 + t_2}}$$

This is a **time-weighted average** of the two rates.
Step 4. Numerically: $r_{\text{eff}} = 6.2/36 \approx 0.172/\text{month}$.

(d) Discrete-to-continuous conversion:
Step 1. Discrete: $N(t+1) = \lambda N(t)$ with $\lambda = 1 + r_d = 1 + 6 = 7$.
Step 2. Continuous equivalent over one generation ($\tau_g = 1$ year): $e^{r_c \cdot 1} = \lambda = 7$.
Step 3. Take natural log of both sides:

$$r_c = \ln 7 \approx 1.946 \text{ yr}^{-1}$$

Step 4. Note $r_c = \ln 7 \neq r_d = 6$. They differ significantly for large r_d (only approximately equal when $r_d \ll 1$).

1.8 Problem 1.8: Gompertz Growth

Problem

Consider Gompertz growth: $\dfrac{dN}{dt} = rN \ln(K/N)$.

(a) Find fixed points and check stability.

(b) Verify $N = K$ is a fixed point.

(c) Compare saturation to logistic: does Gompertz approach K faster or slower near the carrying capacity?

(d) For early times $N \ll K$, what does the equation simplify to?

Solution

(a) Fixed points:
Step 1. Set $\dot{N} = 0$:

$$rN \ln(K/N) = 0$$

Step 2. Apply zero-product: $N = 0$ or $\ln(K/N) = 0$.
Step 3. Solve $\ln(K/N) = 0$:

$$K/N = e^0 = 1 \quad \Longrightarrow \quad N = K$$

Fixed points: $N^* = 0$ (trivial) and $N^* = K$ (carrying capacity).
Step 4. Stability via $f'(N)$ where $f = rN \ln(K/N)$.
Expand: $f = rN(\ln K - \ln N)$. Differentiate using product rule:

$$f'(N) = r(\ln K - \ln N) + rN \cdot \frac{-1}{N} = r \ln(K/N) - r$$

Step 5. At $N^* = K$: $f'(K) = r \ln(K/K) - r = r \cdot 0 - r = -r < 0 \Rightarrow$ **stable**.
Step 6. At $N^* = 0$: $\ln(K/N) \to \infty$ as $N \to 0^+$, so $f'(0^+) \to +\infty \Rightarrow$ **unstable**.

(b) Verify $N = K$:
Step 1. Substitute $N = K$:

$$\dot{N} = rK \ln(K/K) = rK \ln(1) = rK \cdot 0 = 0 \checkmark$$

(c) Speed of approach to K vs logistic:
Step 1. Near K, write $N = K - \delta$ for small $\delta > 0$ (below K):
For **Gompertz**: $\dot{N} = r(K - \delta) \ln(K/(K - \delta)) = r(K - \delta) \ln(1/(1 - \delta/K))$. For small δ/K: $\ln(1/(1 - x)) \approx x$ for $x \ll 1$:

$$\dot{N} \approx r(K - \delta) \cdot \frac{\delta}{K} \approx r\delta \quad (\text{for } \delta \ll K)$$

For **Logistic**: $\dot{N} = r(K - \delta)(1 - (K - \delta)/K) = r(K - \delta) \cdot \delta/K \approx r\delta$.
Step 2. Both give $\dot{N} \approx r\delta$ near K. Same rate of approach to leading order. However the sub-leading terms differ: Gompertz has an additional factor making it **slightly slower** to approach K far from it but similar very close.

(d) Early times $N \ll K$:
Step 1. When $N \ll K$: $K/N \gg 1$, so $\ln(K/N) = \ln K - \ln N$. For very small N, $\ln K$ dominates:

$$\dot{N} \approx rN \ln K$$

Step 2. This is exponential growth with effective rate $r_{\text{eff}} = r \ln K$.
Step 3. Compare to logistic: $\dot{N} \approx rN$ for logistic at small N. Gompertz grows faster early ($r \ln K > r$ when $K > e$), but both are exponential.

2. Predator-Prey Systems

2.1 Problem 2.1: Standard Lotka-Volterra (Foundation)

Problem

Standard equations:

$$\frac{dN}{dt} = rN - \bar{\beta}NP, \qquad \frac{dP}{dt} = \varepsilon\bar{\beta}NP - ZP$$

(a) Find all fixed points.

(b) Compute the interaction matrix A.

(c) Compute the standard Jacobian J^* at the non-trivial fixed point.

(d) Find eigenvalues both ways.

(e) Interpret stability.

Solution

(a) Fixed points:

Step 1. Set $\dot{N} = 0$:

$$rN - \bar{\beta}NP = 0$$

Step 2. Factor N:

$$N(r - \bar{\beta}P) = 0$$

Step 3. Apply zero-product:

$$N = 0 \quad \text{OR} \quad r - \bar{\beta}P = 0 \quad \Longrightarrow \quad P = r/\bar{\beta}$$

(solving $r - \bar{\beta}P = 0$ for P: add $\bar{\beta}P$ then divide by $\bar{\beta}$)

Step 4. Set $\dot{P} = 0$:

$$\varepsilon\bar{\beta}NP - ZP = 0$$

Step 5. Factor P:

$$P(\varepsilon\bar{\beta}N - Z) = 0$$

Step 6. Apply zero-product:

$$P = 0 \quad \text{OR} \quad \varepsilon\bar{\beta}N - Z = 0 \quad \Longrightarrow \quad N = Z/(\varepsilon\bar{\beta})$$

Step 7. Combine:

$$\boxed{(N^*, P^*) = (0, 0) \text{ trivial}, \quad (N^*, P^*) = \left(\frac{Z}{\varepsilon\bar{\beta}}, \frac{r}{\bar{\beta}}\right) \text{ coexistence}}$$

(b) Interaction matrix:

Step 1. Divide $\dot{N}$ by N:

$$\frac{1}{N}\frac{dN}{dt} = \frac{rN}{N} - \frac{\bar{\beta}NP}{N} = r - \bar{\beta}P$$

(cancel one N from each term)

Step 2. Divide $\dot{P}$ by P:

$$\frac{1}{P}\frac{dP}{dt} = \frac{\varepsilon\bar{\beta}NP}{P} - \frac{ZP}{P} = \varepsilon\bar{\beta}N - Z$$

Step 3. Read off coefficients of state variables:

- In $r - \bar{\beta}P$: coefficient of N is 0; coefficient of P is $-\bar{\beta}$.
- In $\varepsilon\bar{\beta}N - Z$: coefficient of N is $\varepsilon\bar{\beta}$; coefficient of P is 0.

Step 4. Build the matrix:

$$A = \begin{pmatrix} 0 & -\bar{\beta} \\ \varepsilon\bar{\beta} & 0 \end{pmatrix}$$

Step 5. Trace:

$$\mathrm{Tr}(A) = 0 + 0 = 0$$

Step 6. Determinant: $ad - bc$ with $a = 0, b = -\bar{\beta}, c = \varepsilon\bar{\beta}, d = 0$:

$$\mathrm{Det}(A) = (0)(0) - (-\bar{\beta})(\varepsilon\bar{\beta}) = 0 - (-\varepsilon\bar{\beta}^2) = \varepsilon\bar{\beta}^2$$

(negative of negative is positive)

(c) Standard Jacobian:
Define $f = rN - \bar{\beta}NP$ and $g = \varepsilon\bar{\beta}NP - ZP$.
Step 1. Compute partial derivatives. Treating P as constant:

$$\frac{\partial f}{\partial N} = \frac{\partial}{\partial N}[rN] - \frac{\partial}{\partial N}[\bar{\beta}NP] = r - \bar{\beta}P$$

($\bar{\beta}P$ is constant w.r.t. N, so derivative of $\bar{\beta}NP$ is $\bar{\beta}P$)
Step 2.

$$\frac{\partial f}{\partial P} = 0 - \bar{\beta}N = -\bar{\beta}N$$

Step 3.

$$\frac{\partial g}{\partial N} = \varepsilon\bar{\beta}P - 0 = \varepsilon\bar{\beta}P$$

Step 4.

$$\frac{\partial g}{\partial P} = \varepsilon\bar{\beta}N - Z$$

Step 5. Build J:

$$J = \begin{pmatrix} r - \bar{\beta}P & -\bar{\beta}N \\ \varepsilon\bar{\beta}P & \varepsilon\bar{\beta}N - Z \end{pmatrix}$$

Step 6. Evaluate at $(N^*, P^*) = (Z/(\varepsilon\bar{\beta}), r/\bar{\beta})$. Entry (1,1):

$$r - \bar{\beta}P^* = r - \bar{\beta} \cdot \frac{r}{\bar{\beta}} = r - r = 0$$

($\bar{\beta}$'s cancel)
Step 7. Entry (1,2):

$$-\bar{\beta}N^* = -\bar{\beta} \cdot \frac{Z}{\varepsilon\bar{\beta}} = -\frac{Z}{\varepsilon}$$

($\bar{\beta}$'s cancel)
Step 8. Entry (2,1):

$$\varepsilon\bar{\beta}P^* = \varepsilon\bar{\beta} \cdot \frac{r}{\bar{\beta}} = \varepsilon r$$

Step 9. Entry (2,2):

$$\varepsilon\bar{\beta}N^* - Z = \varepsilon\bar{\beta} \cdot \frac{Z}{\varepsilon\bar{\beta}} - Z = Z - Z = 0$$

Step 10.

$$\boxed{J^* = \begin{pmatrix} 0 & -Z/\varepsilon \\ \varepsilon r & 0 \end{pmatrix}}$$

(d) Eigenvalues:
From A: $\mathrm{Tr} = 0$, $\mathrm{Det} = \varepsilon\bar{\beta}^2$. Apply formula:

$$\lambda_\pm = \frac{0 \pm \sqrt{0^2 - 4\varepsilon\bar{\beta}^2}}{2} = \frac{\pm\sqrt{-4\varepsilon\bar{\beta}^2}}{2}$$

Step 1. Pull negative out as i:

$$\sqrt{-4\varepsilon\bar{\beta}^2} = i\sqrt{4\varepsilon\bar{\beta}^2} = 2i\bar{\beta}\sqrt{\varepsilon}$$

($\sqrt{-1} = i$, $\sqrt{4\varepsilon\bar{\beta}^2} = 2\bar{\beta}\sqrt{\varepsilon}$)

Step 2.

$$\lambda_\pm = \frac{\pm 2i\bar{\beta}\sqrt{\varepsilon}}{2} = \pm i\bar{\beta}\sqrt{\varepsilon}$$

From J^*: Tr $= 0$, Det $= (0)(0) - (-Z/\varepsilon)(\varepsilon r)$:

$$\mathrm{Det}(J^*) = 0 - (-Zr) = Zr$$

($(-Z/\varepsilon)(\varepsilon r) = -Zr$ since ε's cancel)

Step 3.

$$\lambda_\pm = \pm i\sqrt{Zr}$$

(e) Interpretation:
Pure imaginary eigenvalues ($\mathrm{Re}(\lambda) = 0$, $\mathrm{Im}(\lambda) \neq 0$) $\Rightarrow$ **neutral oscillations / closed limit cycles (center)**. Not strictly stable nor unstable. Real LV populations oscillate forever about the coexistence point with amplitude set by initial conditions.
Why two different magnitudes? A matrix uses scaled coordinates ($\dot{X}_i/X_i$), while J^* uses unscaled. The eigenvalue *magnitudes* differ ($\bar{\beta}\sqrt{\varepsilon}$ vs. $\sqrt{Zr}$), but the *signs of real parts* (and hence stability) always agree.

2.2 Problem 2.2: Type II Functional Response (2024 Q2)

Problem

Replace constant $\bar{\beta}$ with Type II:

$$\bar{\beta} \to \bar{\beta}(N) = \frac{\alpha N}{1 + \alpha h_T N}$$

where α = attack rate, h_T = handling time.

(a) Compute the Jacobian (express in terms of $\bar{\beta}$ where possible).

(b) Evaluate at the unmodified LV fixed point $(N^*, P^*) = (Z/(\varepsilon\bar{\beta}), r/\bar{\beta})$.

(c) Find eigenvalues; compare to LV result.

(d) Why does the system become *less* stable with Type II?

(e) Can we write an interaction matrix A? Why or why not?

Solution

(a) Compute Jacobian:
The equations are:

$$\frac{dN}{dt} = rN - \frac{\alpha NP}{1 + \alpha h_T N}, \qquad \frac{dP}{dt} = \varepsilon\frac{\alpha NP}{1 + \alpha h_T N} - ZP$$

Define $h(N) = \frac{\alpha N}{1+\alpha h_T N}$ so the consumption is $h(N)P$.
Step 1. Compute $h'(N)$ using the quotient rule. Let $u = \alpha N$, $v = 1 + \alpha h_T N$. Then $u' = \alpha$, $v' = \alpha h_T$:

$$h'(N) = \frac{u'v - uv'}{v^2} = \frac{\alpha(1 + \alpha h_T N) - \alpha N \cdot \alpha h_T}{(1 + \alpha h_T N)^2}$$

Step 2. Expand the numerator:

$$\alpha(1 + \alpha h_T N) - \alpha^2 h_T N = \alpha + \alpha^2 h_T N - \alpha^2 h_T N = \alpha$$

($\alpha^2 h_T N$ terms cancel)

Step 3. So:

$$h'(N) = \frac{\alpha}{(1+\alpha h_T N)^2}$$

Step 4. Define $C = 1 + \alpha h_T N$ for shorthand. Then $h(N) = \alpha N/C$ and $h'(N) = \alpha/C^2$.
Step 5. Compute partial derivatives. For $f = rN - h(N)P$:

$$\frac{\partial f}{\partial N} = r - h'(N)P = r - \frac{\alpha P}{C^2}$$

(P is constant w.r.t. N)
Step 6.

$$\frac{\partial f}{\partial P} = -h(N) = -\frac{\alpha N}{C}$$

Step 7. For $g = \varepsilon h(N)P - ZP$:

$$\frac{\partial g}{\partial N} = \varepsilon h'(N)P = \frac{\varepsilon \alpha P}{C^2}$$

Step 8.

$$\frac{\partial g}{\partial P} = \varepsilon h(N) - Z = \frac{\varepsilon \alpha N}{C} - Z$$

Step 9. Build J:

$$J = \begin{pmatrix} r - \alpha P/C^2 & -\alpha N/C \\ \varepsilon \alpha P/C^2 & \varepsilon \alpha N/C - Z \end{pmatrix}$$

(b) Evaluate at the FP $(N^*, P^*) = (Z/(\varepsilon\bar{\beta}), r/\bar{\beta})$:
Here $\bar{\beta} = h(N^*) = \alpha N^*/C^*$ where $C^* = 1 + \alpha h_T N^*$.
Step 1. Useful relations: from $\bar{\beta} = \alpha N^*/C^*$, we get $\alpha N^* = \bar{\beta} C^*$, so $\alpha = \bar{\beta} C^*/N^*$.
Step 2. Entry (1,1): $r - \alpha P^*/(C^*)^2$. Substitute $P^* = r/\bar{\beta}$:

$$r - \frac{\alpha P^*}{(C^*)^2} = r - \frac{\alpha}{(C^*)^2} \cdot \frac{r}{\bar{\beta}}$$

Step 3. Substitute $\alpha = \bar{\beta} C^*/N^*$:

$$= r - \frac{\bar{\beta} C^*/N^*}{(C^*)^2} \cdot \frac{r}{\bar{\beta}} = r - \frac{r}{N^* C^*}$$

($\bar{\beta}$ cancels, $C^*/(C^*)^2 = 1/C^*$)
Step 4. Use $\bar{\beta} = \alpha N^*/C^*$ rearranged: $1/(N^* C^*) = \alpha/(C^*)^2/\bar{\beta} = \alpha/(C^*)^2$ ... Actually a cleaner approach: at the FP, the consumption equation $\varepsilon h(N^*) = Z$ is automatically satisfied (since $\dot{P} = 0$), giving $\varepsilon\bar{\beta} = Z/N^*$. So $\bar{\beta} = Z/(\varepsilon N^*)$.
Step 5. Final form for entry (1,1) using $C \equiv C^*$:

$$r - \frac{r}{N^* C} = r\left(1 - \frac{1}{N^* C}\right)$$

For typical parameter values this is **positive** (in standard LV without Type II, this entry was 0).
Step 6. Entry (1,2): $-\alpha N^*/C = -h(N^*) = -\bar{\beta}$, then multiply $-\bar{\beta}\cdot$?... actually we need $-\bar{\beta}$ but in terms of N^*:

$$-\frac{\alpha N^*}{C^*} = -\bar{\beta}$$

This isn't simplified further.
Step 7. Entry (2,1): $\varepsilon \alpha P^*/(C^*)^2$:

$$\frac{\varepsilon \alpha P^*}{(C^*)^2} = \frac{\varepsilon}{(C^*)^2} \cdot \alpha \cdot \frac{r}{\bar{\beta}} = \frac{\varepsilon r \alpha}{\bar{\beta}(C^*)^2}$$

Substitute $\alpha/\bar{\beta} = (C^*)^2/(N^*C^*) = C^*/N^*$ ... wait this isn't right either. Let me use $\alpha N^* = \bar{\beta}C^*$ so $\alpha/\bar{\beta} = C^*/N^*$:

$$= \frac{\varepsilon r}{(C^*)^2} \cdot \frac{C^*}{N^*} \cdot \bar{\beta} \cdot \frac{1}{\bar{\beta}} = \frac{\varepsilon r}{N^*C^*}$$

Hmm, this is getting algebraically tangled. Let me just keep entries symbolic.
Step 8. Entry (2,2): $\varepsilon h(N^*) - Z = \varepsilon\bar{\beta} - Z = 0$ at the FP (this IS the FP condition, since dividing $\dot{P} = 0$ by P gives this).
So:

$$J^* = \begin{pmatrix} r(1 - 1/(N^*C)) & -\bar{\beta} \\ \cdot & 0 \end{pmatrix}$$

where (2,1) entry is some positive quantity.

(c) Eigenvalues qualitatively:
Step 1. Trace:

$$\mathrm{Tr}(J^*) = r\left(1 - \frac{1}{N^*C}\right) + 0$$

Step 2. For typical parameters $N^*C > 1$, so Tr > 0.
Step 3. Compare to standard LV: Tr was **0** (neutral oscillations). Now Tr > 0.
Step 4. A positive trace means **at least one eigenvalue has positive real part**. Combined with positive Det, the FP is an **unstable spiral**.

(d) Why Type II destabilizes:
The constant-rate $\bar{\beta}$ assumed predators can consume prey at any density — the more prey, the more they eat, no saturation. Type II adds saturation: with handling time h_T, predator consumption levels off at high prey density. When prey become abundant, predators can't keep up, and prey "escape" control. This means small perturbations away from the FP grow (oscillations gain amplitude), whereas standard LV had only neutral cycles. In the full nonlinear system, the destabilization typically saturates into a stable limit cycle.

(e) Why no A matrix:
Step 1. For the A matrix we'd compute $\dot{N}/N$:

$$\frac{\dot{N}}{N} = r - \frac{\alpha P}{1 + \alpha h_T N}$$

Step 2. The coefficient of P here is $-\alpha/(1 + \alpha h_T N)$, which depends on N. This is **not** the constant-times-state-variable form $A_{ij}X_j$ required for the interaction matrix.
Step 3. Conclusion: **cannot** write A. The Type II response inherently breaks the bilinear $A_{ij}X_iX_j$ structure that the interaction matrix method requires.

2.3 Problem 2.3: Pack Hunting (2024 Q1)

Problem

4 predators always hunt in packs to kill 1 prey at a time.

(a) Choose appropriate compartments.

(b) Write the differential equations.

(c) Interpret parameters relative to standard LV.

(d) Write A, find Tr, Det, eigenvalues.

(e) Is this system more or less stable?

Solution

(a) Compartments: Let N = prey density, P_p = density of **packs** (not individual predators). One pack contains 4 predators.
(b) Equations: A pack consumes prey at rate $\bar{\beta}_p N P_p$. Each kill feeds the pack:

$$\frac{dN}{dt} = rN - \bar{\beta}_p N P_p$$

$$\frac{dP_p}{dt} = \varepsilon \bar{\beta}_p N P_p - Z_p P_p$$

Same structure as standard LV; only the units of P_p and consumption rate change.
(c) Parameters:

- r: same units (prey/time per prey).
- $\bar{\beta}_p$: consumption rate per pack. Since each pack eats as much as 4 individual predators in coordinated kills, $\bar{\beta}_p \approx 4\bar{\beta}$ when comparing to individual predators.
- ε: still dimensionless, prey-biomass-to-predator-biomass conversion. Same value.
- Z_p: pack mortality. Hard to define exactly — average pack persistence rate. Roughly same magnitude as Z.

(d) Interaction matrix:

$$A = \begin{pmatrix} 0 & -\bar{\beta}_p \\ \varepsilon \bar{\beta}_p & 0 \end{pmatrix}$$

$\mathrm{Tr}(A) = 0$, $\mathrm{Det}(A) = \varepsilon \bar{\beta}_p^2$. Eigenvalues:

$$\lambda_\pm = \pm i \bar{\beta}_p \sqrt{\varepsilon}$$

(e) Stability: Eigenvalues are pure imaginary $\Rightarrow$ neutral limit cycles (same conclusion as LV). However, since $\bar{\beta}_p \approx 4\bar{\beta}$, the magnitude $|\lambda|$ is $4\times$ larger, meaning **larger-amplitude oscillations**. The pack system has bigger boom-bust cycles, getting closer to extinction in the troughs. Strictly: same stability class. Loosely: **less stable in practice** due to extinction risk.

2.4 Problem 2.4: Cannibalistic Predator (2021 Q2)

Problem

Modify LV for cannibalistic predator:

$$\frac{dN}{dt} = rN - \bar{\beta} N P, \quad \frac{dP}{dt} = \varepsilon \bar{\beta} N P - (1-\varepsilon)\bar{\beta}_c P^2 - ZP$$

(a) Draw the interaction network.

(b) Write A. Find Tr and Det.

(c) Find fixed points.

(d) Compute J^* at non-trivial FP.

(e) Find eigenvalues. What happens when $\bar{\beta}_c \gg \bar{\beta}$?

Solution

(a) Network diagram: Two nodes (prey N, predator P). Arrow from N to P representing predation. **Self-loop on** P representing cannibalism (predator eats predator).

(b) Interaction matrix:
Step 1. Divide each equation by its variable. For $\dot{N}$:

$$\frac{1}{N}\frac{dN}{dt} = r - \bar{\beta} P$$

Step 2. For $\dot{P}$:

$$\frac{1}{P}\frac{dP}{dt} = \varepsilon\bar{\beta}N - (1-\varepsilon)\bar{\beta}_c P - Z$$

(cancel one P from P^2 term to get $-(1-\varepsilon)\bar{\beta}_c P$)

Step 3. Read coefficients:

- In $r - \bar{\beta}P$: coeff of N is 0; coeff of P is $-\bar{\beta}$.
- In $\varepsilon\bar{\beta}N - (1-\varepsilon)\bar{\beta}_c P - Z$: coeff of N is $\varepsilon\bar{\beta}$; coeff of P is $-(1-\varepsilon)\bar{\beta}_c$.

Step 4.

$$A = \begin{pmatrix} 0 & -\bar{\beta} \\ \varepsilon\bar{\beta} & -(1-\varepsilon)\bar{\beta}_c \end{pmatrix}$$

Step 5. Trace: sum of diagonal:

$$\mathrm{Tr}(A) = 0 + [-(1-\varepsilon)\bar{\beta}_c] = -(1-\varepsilon)\bar{\beta}_c$$

Since $0 < \varepsilon < 1$ and $\bar{\beta}_c > 0$: $\mathbf{Tr}(A) < 0$ — **stabilizing!**

Step 6. Determinant:

$$\mathrm{Det}(A) = (0)\cdot[-(1-\varepsilon)\bar{\beta}_c] - (-\bar{\beta})(\varepsilon\bar{\beta}) = 0 + \varepsilon\bar{\beta}^2 = \varepsilon\bar{\beta}^2 > 0$$

(c) Fixed points:

Step 1. Set $\dot{N} = 0$:

$$N(r - \bar{\beta}P) = 0 \quad \Longrightarrow \quad N = 0 \text{ or } P = r/\bar{\beta}$$

Step 2. Set $\dot{P} = 0$:

$$\varepsilon\bar{\beta}NP - (1-\varepsilon)\bar{\beta}_c P^2 - ZP = 0$$

Step 3. Factor P:

$$P[\varepsilon\bar{\beta}N - (1-\varepsilon)\bar{\beta}_c P - Z] = 0$$

Step 4. Either $P = 0$ or $\varepsilon\bar{\beta}N - (1-\varepsilon)\bar{\beta}_c P - Z = 0$. Solve the second:

$$\varepsilon\bar{\beta}N = (1-\varepsilon)\bar{\beta}_c P + Z$$

Step 5. Solve for N:

$$N = \frac{(1-\varepsilon)\bar{\beta}_c P + Z}{\varepsilon\bar{\beta}}$$

Step 6. Substitute $P^* = r/\bar{\beta}$:

$$N^* = \frac{(1-\varepsilon)\bar{\beta}_c \cdot (r/\bar{\beta}) + Z}{\varepsilon\bar{\beta}}$$

Step 7. Get common denominator in numerator: $(1-\varepsilon)\bar{\beta}_c r/\bar{\beta} + Z = [(1-\varepsilon)\bar{\beta}_c r + Z\bar{\beta}]/\bar{\beta}$:

$$N^* = \frac{(1-\varepsilon)\bar{\beta}_c r + Z\bar{\beta}}{\bar{\beta}\cdot\varepsilon\bar{\beta}} = \frac{Z\bar{\beta} + (1-\varepsilon)\bar{\beta}_c r}{\varepsilon\bar{\beta}^2}$$

(multiply both sides by $\bar{\beta}$, then the $\bar{\beta}\cdot\varepsilon\bar{\beta} = \varepsilon\bar{\beta}^2$ in denominator)

$$\boxed{N^* = \frac{Z\bar{\beta} + (1-\varepsilon)\bar{\beta}_c r}{\varepsilon\bar{\beta}^2}, \quad P^* = \frac{r}{\bar{\beta}}}$$

(d) Standard Jacobian:

Step 1. Define $f = rN - \bar{\beta}NP$ and $g = \varepsilon\bar{\beta}NP - (1-\varepsilon)\bar{\beta}_c P^2 - ZP$.

Step 2. Partials of f (same as standard LV):

$$\frac{\partial f}{\partial N} = r - \bar{\beta}P, \quad \frac{\partial f}{\partial P} = -\bar{\beta}N$$

Step 3. Partial of g w.r.t. N:

$$\frac{\partial g}{\partial N} = \varepsilon\bar{\beta}P - 0 - 0 = \varepsilon\bar{\beta}P$$

Step 4. Partial of g w.r.t. P. Note $\partial[P^2]/\partial P = 2P$:

$$\frac{\partial g}{\partial P} = \varepsilon\bar{\beta}N - 2(1-\varepsilon)\bar{\beta}_c P - Z$$

Step 5. Build Jacobian:

$$J = \begin{pmatrix} r - \bar{\beta}P & -\bar{\beta}N \\ \varepsilon\bar{\beta}P & \varepsilon\bar{\beta}N - 2(1-\varepsilon)\bar{\beta}_c P - Z \end{pmatrix}$$

Step 6. Evaluate at FP. Entry (1,1):

$$r - \bar{\beta}P^* = r - \bar{\beta}\cdot\frac{r}{\bar{\beta}} = r - r = 0$$

Step 7. Entry (1,2): $-\bar{\beta}N^*$ (keep symbolic).
Step 8. Entry (2,1):

$$\varepsilon\bar{\beta}P^* = \varepsilon\bar{\beta}\cdot\frac{r}{\bar{\beta}} = \varepsilon r$$

Step 9. Entry (2,2). Use the FP condition $\varepsilon\bar{\beta}N^* - (1-\varepsilon)\bar{\beta}_c P^* - Z = 0$, which gives $\varepsilon\bar{\beta}N^* - Z = (1-\varepsilon)\bar{\beta}_c P^*$:

$$\varepsilon\bar{\beta}N^* - 2(1-\varepsilon)\bar{\beta}_c P^* - Z = [(1-\varepsilon)\bar{\beta}_c P^*] - 2(1-\varepsilon)\bar{\beta}_c P^*$$

Step 10. Simplify: $X - 2X = -X$:

$$= -(1-\varepsilon)\bar{\beta}_c P^*$$

Step 11. Substitute $P^* = r/\bar{\beta}$:

$$= -(1-\varepsilon)\bar{\beta}_c\cdot\frac{r}{\bar{\beta}} = -\frac{(1-\varepsilon)\bar{\beta}_c r}{\bar{\beta}}$$

Step 12. Final J^*:

$$\boxed{J^* = \begin{pmatrix} 0 & -\bar{\beta}N^* \\ \varepsilon r & -(1-\varepsilon)\bar{\beta}_c r/\bar{\beta} \end{pmatrix}}$$

(e) Eigenvalues:
Step 1. Trace:

$$\mathrm{Tr}(J^*) = 0 + \left[-\frac{(1-\varepsilon)\bar{\beta}_c r}{\bar{\beta}}\right] = -\frac{(1-\varepsilon)\bar{\beta}_c r}{\bar{\beta}} < 0$$

Step 2. Determinant: $ad - bc$ with $a = 0$, $b = -\bar{\beta}N^*$, $c = \varepsilon r$, $d = -(1-\varepsilon)\bar{\beta}_c r/\bar{\beta}$:

$$\mathrm{Det}(J^*) = (0)\cdot d - (-\bar{\beta}N^*)(\varepsilon r) = 0 + \varepsilon r\bar{\beta}N^* = \varepsilon r\bar{\beta}N^* > 0$$

Step 3. Apply formula:

$$\lambda_\pm = \frac{\mathrm{Tr}}{2}\left(1 \pm \sqrt{1 - \frac{4\mathrm{Det}}{\mathrm{Tr}^2}}\right)$$

Step 4. $\mathrm{Tr} < 0$ and $\mathrm{Det} > 0$ guarantee $\mathrm{Re}(\lambda_\pm) < 0$ regardless of discriminant sign $\Rightarrow$ **stable**.
Step 5. Discriminant $\Delta = 1 - 4\mathrm{Det}/\mathrm{Tr}^2$. Substitute:

$$\Delta = 1 - \frac{4\varepsilon r\bar{\beta}N^*}{[(1-\varepsilon)\bar{\beta}_c r/\bar{\beta}]^2} = 1 - \frac{4\varepsilon r\bar{\beta}N^*\bar{\beta}^2}{(1-\varepsilon)^2\bar{\beta}_c^2 r^2}$$

Step 6. Simplify: $4\varepsilon\bar{\beta}^3 N^*/((1-\varepsilon)^2\bar{\beta}_c^2 r)$:

$$\Delta = 1 - \frac{4\varepsilon\bar{\beta}^3 N^*}{(1-\varepsilon)^2\bar{\beta}_c^2 r}$$

Step 7. Limit $\bar{\beta}_c \gg \bar{\beta}$: The fraction in Δ scales as $1/\bar{\beta}_c^2$, which goes to 0. So $\Delta \to 1 > 0$.
Step 8. Positive discriminant $\Rightarrow$ real eigenvalues:

$$\lambda_\pm = \frac{\mathrm{Tr}}{2}(1 \pm \sqrt{\Delta})$$

Both negative since $\mathrm{Tr} < 0$. **No oscillations: stable node.**
Conclusion: Strong cannibalism eliminates the boom-bust cycles entirely. Predators self-buffer against prey scarcity by eating each other.

2.5 Problem 2.5: Three-Species System (2023 Q2)

Problem

Pet ownership system: humans H, pets D, predators P.

$$\frac{dH}{dt} = rH(1 - H/K) + \alpha_D HD$$

$$\frac{dD}{dt} = rD(1 - D/K) + \omega\alpha_D HD - \bar{\beta}DP$$

$$\frac{dP}{dt} = \varepsilon\bar{\beta}DP - ZP$$

(a) How does this differ from standard predator-prey?

(b) Write the interaction matrix A.

(c) Find fixed points.

(d) Compute Jacobian at coexistence FP.

Solution

(a) Differences from standard predator-prey:

- **Three compartments** instead of two (H, D, P).
- **Mutualism between H and D**: term $+\alpha_D HD$ appears in BOTH $\dot{H}$ and (with weight ω) $\dot{D}$. Both benefit from contact.
- **Asymmetric weights**: parameter ω allows different benefits (H vs D).
- **Logistic growth** in both H and D (carrying capacity K each).
- Compartments are concrete species (humans, pets, predators), not abstractions.

(b) Interaction matrix:
Step 1. Divide each equation by its variable. For $\dot{H}$:

$$\frac{1}{H}\frac{dH}{dt} = r(1 - H/K) + \alpha_D D$$

Step 2. Distribute r:

$$= r - \frac{rH}{K} + \alpha_D D$$

Step 3. For $\dot{D}$:

$$\frac{1}{D}\frac{dD}{dt} = r - \frac{rD}{K} + \omega\alpha_D H - \bar{\beta}P$$

Step 4. For $\dot{P}$:

$$\frac{1}{P}\frac{dP}{dt} = \varepsilon\bar{\beta}D - Z$$

Step 5. Read coefficients of state variables (H, D, P) in each per-capita rate:

- $\dot{H}/H$: H coeff $= -r/K$, D coeff $= \alpha_D$, P coeff $= 0$.
- $\dot{D}/D$: H coeff $= \omega\alpha_D$, D coeff $= -r/K$, P coeff $= -\bar{\beta}$.
- $\dot{P}/P$: H coeff $= 0$, D coeff $= \varepsilon\bar{\beta}$, P coeff $= 0$.

Step 6.

$$\boxed{A = \begin{pmatrix} -r/K & \alpha_D & 0 \\ \omega\alpha_D & -r/K & -\bar{\beta} \\ 0 & \varepsilon\bar{\beta} & 0 \end{pmatrix}}$$

Step 7. Trace:

$$\mathrm{Tr}(A) = -r/K + (-r/K) + 0 = -2r/K < 0 \quad \text{(stabilizing)}$$

(c) Fixed points:
Step 1. At a non-trivial FP, set each per-capita rate to zero (since each $X^* \neq 0$). Three equations:

$$(\text{eq } 1) : r - rH/K + \alpha_D D = 0$$

$$(\text{eq } 2) : r - rD/K + \omega\alpha_D H - \bar{\beta}P = 0$$

$$(\text{eq } 3) : \varepsilon\bar{\beta}D - Z = 0$$

Step 2. Solve eq 3 for D:

$$\varepsilon\bar{\beta}D = Z \quad \Longrightarrow \quad D^* = \frac{Z}{\varepsilon\bar{\beta}}$$

Step 3. Substitute into eq 1:

$$r - rH^*/K + \alpha_D \cdot \frac{Z}{\varepsilon\bar{\beta}} = 0$$

Step 4. Solve for H^*:

$$\frac{rH^*}{K} = r + \frac{\alpha_D Z}{\varepsilon\bar{\beta}}$$

$$H^* = K + \frac{K\alpha_D Z}{r\varepsilon\bar{\beta}} = K\left(1 + \frac{\alpha_D Z}{r\varepsilon\bar{\beta}}\right)$$

(multiply both sides by K/r, then factor K)
Step 5. Substitute D^* and H^* into eq 2 to find P^*:

$$P^* = \frac{r - rD^*/K + \omega\alpha_D H^*}{\bar{\beta}}$$

(Symbolic; positive if $\omega\alpha_D H^* > rD^*/K - r$.)

(d) Jacobian at the coexistence FP:
Step 1. Define $f_1 = rH(1 - H/K) + \alpha_D HD$ etc. Compute partials.
Step 2. For $f_1 = rH - rH^2/K + \alpha_D HD$:

$$\partial f_1/\partial H = r - 2rH/K + \alpha_D D$$

$$\partial f_1/\partial D = \alpha_D H, \qquad \partial f_1/\partial P = 0$$

Step 3. For $f_2 = rD - rD^2/K + \omega\alpha_D HD - \bar{\beta}DP$:

$$\partial f_2/\partial H = \omega\alpha_D D$$

$$\partial f_2/\partial D = r - 2rD/K + \omega\alpha_D H - \bar{\beta}P$$

$$\partial f_2/\partial P = -\bar{\beta}D$$

Step 4. For $f_3 = \varepsilon\bar{\beta}DP - ZP$:

$$\partial f_3/\partial H = 0, \quad \partial f_3/\partial D = \varepsilon\bar{\beta}P, \quad \partial f_3/\partial P = \varepsilon\bar{\beta}D - Z$$

Step 5. Build J:

$$J = \begin{pmatrix} r - 2rH/K + \alpha_D D & \alpha_D H & 0 \\ \omega\alpha_D D & r - 2rD/K + \omega\alpha_D H - \bar{\beta}P & -\bar{\beta}D \\ 0 & \varepsilon\bar{\beta}P & \varepsilon\bar{\beta}D - Z \end{pmatrix}$$

Step 6. Simplify (1,1) at the FP. Use eq 1 ($r + \alpha_D D^* = rH^*/K$):

$$r - 2rH^*/K + \alpha_D D^* = (r + \alpha_D D^*) - 2rH^*/K = rH^*/K - 2rH^*/K = -rH^*/K$$

Step 7. Similarly (2,2): use eq 2 ($r + \omega\alpha_D H^* - \bar{\beta}P^* = rD^*/K$):

$$r - 2rD^*/K + \omega\alpha_D H^* - \bar{\beta}P^* = rD^*/K - 2rD^*/K = -rD^*/K$$

Step 8. (3,3) entry: from eq 3, $\varepsilon\bar{\beta}D^* - Z = 0$.
Step 9. Final J^*:

$$J^* = \begin{pmatrix} -rH^*/K & \alpha_D H^* & 0 \\ \omega\alpha_D D^* & -rD^*/K & -\bar{\beta}D^* \\ 0 & \varepsilon\bar{\beta}P^* & 0 \end{pmatrix}$$

Step 10. Trace:

$$\mathrm{Tr}(J^*) = -rH^*/K - rD^*/K + 0 = -\frac{r(H^* + D^*)}{K} < 0$$

Step 11. Determinant. Expand along first column ($a_{11} = -rH^*/K$, then 0, 0):

$$\mathrm{Det}(J^*) = -\frac{rH^*}{K} \det \begin{pmatrix} -rD^*/K & -\bar{\beta}D^* \\ \varepsilon\bar{\beta}P^* & 0 \end{pmatrix} - 0 + 0$$

Step 12. Compute the 2×2 sub-det: $(-rD^*/K)(0) - (-\bar{\beta}D^*)(\varepsilon\bar{\beta}P^*) = 0 + \varepsilon\bar{\beta}^2 D^* P^*$:

$$\mathrm{Det}(J^*) = -\frac{rH^*}{K} \cdot \varepsilon\bar{\beta}^2 D^* P^* < 0$$

Step 13. For 3×3 stability via Routh-Hurwitz: characteristic polynomial $\lambda^3 + a_2\lambda^2 + a_1\lambda + a_0 = 0$ with $a_2 = -\mathrm{Tr}(J^*) > 0$, $a_0 = -\mathrm{Det}(J^*) > 0$, and need $a_2 a_1 > a_0$.
Step 14. Both Tr < 0 and Det < 0 have correct sign for stability. Combined with sufficient logistic damping (large r/K), the third Routh-Hurwitz condition is satisfied. **Likely stable.**

2.6 Problem 2.6: LV with Logistic Prey (2017 Q2)

Problem

Modify standard LV by adding logistic self-regulation to the prey:

$$\frac{dN}{dt} = rN - \frac{r}{K}N^2 - \bar{\beta}NP, \qquad \frac{dP}{dt} = \varepsilon\bar{\beta}NP - ZP$$

(a) Write the interaction matrix A and explain how it differs from standard LV.

(b) Find the non-trivial fixed point (N^*, P^*).

(c) Compute $\mathrm{Tr}(A)$ and $\mathrm{Det}(A)$. Are eigenvalues real or complex?

(d) Classify stability. Is this more or less stable than standard LV?

Solution

(a) Interaction matrix:

Step 1. Divide $\dot{N}$ by N:

$$\frac{\dot{N}}{N} = r - \frac{r}{K}N - \bar{\beta}P$$

(cancel one N from rN^2/K to get rN/K)

Step 2. Divide $\dot{P}$ by P:

$$\frac{\dot{P}}{P} = \varepsilon\bar{\beta}N - Z$$

Step 3. Read off coefficients of N and P:

- $\dot{N}/N$: coeff of N is $-r/K$ (self-interaction), coeff of P is $-\bar{\beta}$.
- $\dot{P}/P$: coeff of N is $\varepsilon\bar{\beta}$, coeff of P is 0.

Step 4.

$$A = \begin{pmatrix} -r/K & -\bar{\beta} \\ \varepsilon\bar{\beta} & 0 \end{pmatrix}$$

Difference from standard LV: The (1,1) entry is $-r/K < 0$ instead of 0. This is the self-regulation (intraspecific competition) from the logistic term.

(b) Fixed points:

Step 1. Set per-capita rates to zero:

$$r - \frac{r}{K}N - \bar{\beta}P = 0 \quad \text{(prey eq)}$$

$$\varepsilon\bar{\beta}N - Z = 0 \quad \Longrightarrow \quad N^* = \frac{Z}{\varepsilon\bar{\beta}}$$

Step 2. Substitute N^* into prey equation:

$$r - \frac{r}{K} \cdot \frac{Z}{\varepsilon\bar{\beta}} - \bar{\beta}P^* = 0$$

Step 3. Solve for P^*:

$$\bar{\beta}P^* = r - \frac{rZ}{K\varepsilon\bar{\beta}} = r\left(1 - \frac{Z}{K\varepsilon\bar{\beta}}\right)$$

$$P^* = \frac{r}{\bar{\beta}}\left(1 - \frac{Z}{K\varepsilon\bar{\beta}}\right) = \frac{r}{\bar{\beta}} - \frac{rZ}{K\varepsilon\bar{\beta}^2}$$

(divide by $\bar{\beta}$)

Step 4. Biological constraint: $P^* > 0$ requires $1 > Z/(K\varepsilon\bar{\beta})$, i.e., prey carrying capacity large enough to sustain predators.

(c) Tr, Det, discriminant:

Step 1. Trace:

$$\mathrm{Tr}(A) = -r/K + 0 = -r/K < 0$$

Step 2. Determinant:

$$\mathrm{Det}(A) = (-r/K)(0) - (-\bar{\beta})(\varepsilon\bar{\beta}) = 0 + \varepsilon\bar{\beta}^2 = \varepsilon\bar{\beta}^2 > 0$$

Step 3. Discriminant $\Delta = \mathrm{Tr}^2 - 4\mathrm{Det}$:

$$\Delta = \frac{r^2}{K^2} - 4\varepsilon\bar{\beta}^2$$

If $\Delta < 0$ (i.e., $4\varepsilon\bar{\beta}^2 > r^2/K^2$, or $K > r/(2\bar{\beta}\sqrt{\varepsilon})$): **complex eigenvalues (spiral)**.
If $\Delta > 0$: **real eigenvalues (node)**.

(d) Stability classification:

Step 1. $\mathrm{Tr} < 0$ and $\mathrm{Det} > 0 \Rightarrow$ **stable** in all cases.

Step 2. If $\Delta < 0$ (common when coupling is strong): **stable spiral** — oscillations that dampen into the equilibrium.

Step 3. If $\Delta > 0$: **stable node** — smooth exponential decay to equilibrium, no oscillations.

Step 4. Comparison with standard LV ($\mathrm{Tr} = 0$, neutral oscillations):

- Adding logistic self-regulation shifts Tr from 0 to $-r/K < 0$.
- This converts neutral (undamped) oscillations into **damped** oscillations.
- The system is **more stable**: perturbations decay rather than persist forever.

2.7 Problem 2.7: Prion Disease as LV (2018 Q2)

Problem

Prion disease: normal protein Y_1 converts to misfolded Y_2 upon contact. Treat analogously to predator-prey:

$$\frac{dY_1}{dt} = rY_1 - \bar{\beta}Y_1Y_2, \qquad \frac{dY_2}{dt} = \varepsilon\bar{\beta}Y_1Y_2 - ZY_2$$

(a) Interpret each parameter in the prion context.

(b) Which parameters might have different values than in a biological predator-prey system? Why?

(c) Find fixed points and compute eigenvalues at the coexistence FP.

(d) What would happen if $r = 0$ (no natural production of Y_1)? Interpret biologically.

Solution

(a) Parameter interpretation:

- r: production rate of normal protein Y_1 (e.g., by ribosomes). Unlike prey, Y_1 doesn't self-replicate by interaction — it's produced continuously.
- $\bar{\beta}$: conversion rate when Y_1 contacts Y_2 (misfolding induction rate). Dimensionless-ish: rate per Y_1 per Y_2 per time.
- ε: conversion efficiency. Each Y_1 consumed creates ε units of Y_2. In prion context $\varepsilon = 1$ exactly (every normal protein that misfolds becomes one prion), unlike biological predation where $\varepsilon < 1$.
- Z: degradation rate of misfolded protein Y_2 (by proteasome, chaperones, etc.).

(b) Parameters with different values:
Step 1. ε: Likely $\varepsilon = 1$ for prions (exact 1-to-1 conversion of Y_1 into Y_2). In predator-prey, $\varepsilon \ll 1$ (inefficient energy conversion).
Step 2. r: In predator-prey, prey reproduce by breeding (proportional to N). Here Y_1 is produced at a constant rate from cellular machinery — so really r is not a per-capita rate but an absolute production rate. The equation should perhaps be $\dot{Y}_1 = r_0 - \bar{\beta}Y_1Y_2$ with constant r_0. Using rY_1 approximates this when Y_1 stays near a baseline.
Step 3. Z: Degradation of Y_2 is very slow in prion diseases (that's why they're lethal) — Z is much smaller than in typical predator-prey.

(c) Fixed points and eigenvalues:
Step 1. Same algebra as standard LV. Non-trivial FP:

$$N^* = Y_1^* = Z/(\varepsilon\bar{\beta}), \quad P^* = Y_2^* = r/\bar{\beta}$$

Step 2. Interaction matrix:

$$A = \begin{pmatrix} 0 & -\bar{\beta} \\ \varepsilon\bar{\beta} & 0 \end{pmatrix}$$

Step 3. Tr $= 0$, Det $= \varepsilon\bar{\beta}^2$. Eigenvalues:

$$\lambda_\pm = \pm i\bar{\beta}\sqrt{\varepsilon}$$

Neutral oscillations. In the prion context: normal and misfolded protein oscillate indefinitely.

(d) What if $r = 0$?
Step 1. With $r = 0$: $\dot{Y}_1 = -\bar{\beta}Y_1Y_2$. The fixed point $Y_2^* = r/\bar{\beta} = 0$ collapses.

Step 2. Equations become:

$$\dot{Y}_1 = -\bar{\beta} Y_1 Y_2, \quad \dot{Y}_2 = \varepsilon \bar{\beta} Y_1 Y_2 - Z Y_2$$

Step 3. Without Y_1 production, the only steady state is $Y_1 = 0$ or $Y_2 = 0$.
Step 4. Biological interpretation: if the cell stops producing normal protein Y_1 (e.g., gene silenced, disease state) then Y_1 will be gradually consumed until it is depleted. Then Y_2 will degrade at rate Z (no more substrate). Long-term: both go to zero. **The disease could self-limit!** This is actually the basis of some prion therapy approaches: reduce Y_1 production to starve Y_2.

2.8 Problem 2.8: Competing Species

Problem

Two species N_1 and N_2 compete for the same resource. Each grows logistically and is harmed by the other:

$$\frac{dN_1}{dt} = r_1 N_1 \left(1 - \frac{N_1}{K_1} - \frac{\alpha_{12}}{K_1} N_2\right)$$

$$\frac{dN_2}{dt} = r_2 N_2 \left(1 - \frac{N_2}{K_2} - \frac{\alpha_{21}}{K_2} N_1\right)$$

(a) Write the interaction matrix A.

(b) Find all four fixed points.

(c) Under what conditions does N_1 win? N_2 win? Coexistence? Bistability?

Solution

(a) Interaction matrix:
Step 1. Divide $\dot{N}_1$ by N_1:

$$\frac{\dot{N}_1}{N_1} = r_1 \left(1 - \frac{N_1}{K_1} - \frac{\alpha_{12} N_2}{K_1}\right) = r_1 - \frac{r_1}{K_1} N_1 - \frac{r_1 \alpha_{12}}{K_1} N_2$$

Step 2. Divide $\dot{N}_2$ by N_2 similarly.
Step 3.

$$A = \begin{pmatrix} -r_1/K_1 & -r_1\alpha_{12}/K_1 \\ -r_2\alpha_{21}/K_2 & -r_2/K_2 \end{pmatrix}$$

(b) Four fixed points:
FP 1: Extinction of both: $(N_1^*, N_2^*) = (0, 0)$.
FP 2: Only N_1 survives ($N_2 = 0$): set $\dot{N}_1/N_1 = 0$ with $N_2 = 0$: $r_1(1 - N_1/K_1) = 0 \Rightarrow N_1^* = K_1$.

$$(K_1, 0)$$

FP 3: Only N_2 survives ($N_1 = 0$): similarly $(0, K_2)$.
FP 4: Coexistence. Solve simultaneously:

$$1 - N_1/K_1 - \alpha_{12} N_2/K_1 = 0$$

$$1 - N_2/K_2 - \alpha_{21} N_1/K_2 = 0$$

From eq 1: $N_1 = K_1 - \alpha_{12} N_2$. Substitute into eq 2:

$$1 - \frac{N_2}{K_2} - \frac{\alpha_{21}(K_1 - \alpha_{12} N_2)}{K_2} = 0$$

$$K_2 - N_2 - \alpha_{21} K_1 + \alpha_{21}\alpha_{12} N_2 = 0$$

$$N_2(1 - \alpha_{21}\alpha_{12}) = K_2 - \alpha_{21} K_1$$

$$N_2^* = \frac{K_2 - \alpha_{21}K_1}{1 - \alpha_{12}\alpha_{21}}, \qquad N_1^* = \frac{K_1 - \alpha_{12}K_2}{1 - \alpha_{12}\alpha_{21}}$$

(c) Competition outcomes:
The key quantity is the product $\alpha_{12}\alpha_{21}$ (mutual inhibition strength).
N_1 **wins (competitive exclusion):** $K_1 > \alpha_{12}K_2$ and $K_2 < \alpha_{21}K_1$. *(N_1 can grow when N_2 is at its carrying capacity, but not vice versa)*
N_2 **wins (competitive exclusion):** $K_2 > \alpha_{21}K_1$ and $K_1 < \alpha_{12}K_2$.
Stable coexistence: $\alpha_{12}\alpha_{21} < 1$ and $N_1^* > 0$, $N_2^* > 0$. *(Intraspecific competition stronger than interspecific; species limit themselves more than each other)*
Bistability (unstable coexistence): $\alpha_{12}\alpha_{21} > 1$. *(Interspecific competition stronger; whichever species is more abundant tends to exclude the other; outcome depends on initial conditions)*
Biological intuition: Competitive exclusion (the principle that two species competing for the same resource cannot stably coexist) is recovered when $\alpha_{12}\alpha_{21} \geq 1$. Coexistence is possible only when each species inhibits itself more than it inhibits the competitor.

3. Disease (SIR / SIRS) Models

3.1 Problem 3.1: Standard SIR (Foundation)

Problem

$$\frac{dS}{dt} = -\beta SI, \quad \frac{dI}{dt} = \beta SI - \gamma I, \quad \frac{dR}{dt} = \gamma I$$

(a) Show $S + I + R$ is conserved.

(b) Find all fixed points.

(c) Compute J^* at the disease-free state ($S^* = N, I^* = 0, R^* = 0$).

(d) Find eigenvalues, define R_0.

(e) When does the disease spread?

Solution

(a) Conservation:
Step 1. Sum all three equations:

$$\frac{dS}{dt} + \frac{dI}{dt} + \frac{dR}{dt} = (-\beta SI) + (\beta SI - \gamma I) + (\gamma I)$$

Step 2. Distribute parentheses:

$$= -\beta SI + \beta SI - \gamma I + \gamma I$$

Step 3. Cancel pairs:

$$= 0 + 0 = 0 \checkmark$$

($-\beta SI + \beta SI = 0$ and $-\gamma I + \gamma I = 0$)
Step 4. So $\frac{d}{dt}(S + I + R) = 0$, meaning $S + I + R = N$ constant. Eliminate $R = N - S - I$.

(b) Fixed points:
Step 1. Set $\dot{S} = 0$:

$$-\beta SI = 0 \quad \Longrightarrow \quad S = 0 \text{ or } I = 0$$

Step 2. Set $\dot{I} = 0$:

$$\beta SI - \gamma I = 0$$

Step 3. Factor I:

$$I(\beta S - \gamma) = 0 \quad \Longrightarrow \quad I = 0 \text{ or } S = \gamma/\beta$$

(solving $\beta S - \gamma = 0$ for S)
Step 4. Combine: **disease-free state** $I^* = 0$, $S^* = N$ (free; together with conservation $R^* = 0$).
Step 5. Endemic possibility: $S^* = \gamma/\beta$, $I^* > 0$. But this requires $\gamma/\beta < N$, i.e., $R_0 = \beta N/\gamma > 1$.

(c) Jacobian at disease-free FP:
Define $f(S, I) = -\beta SI$, $g(S, I) = \beta SI - \gamma I$. (Eliminated R via conservation.)
Step 1. Partials of f:

$$\frac{\partial f}{\partial S} = \frac{\partial}{\partial S}[-\beta SI] = -\beta I$$

(I is constant w.r.t. S)
Step 2.

$$\frac{\partial f}{\partial I} = -\beta S$$

Step 3. Partials of g:

$$\frac{\partial g}{\partial S} = \beta I$$

Step 4.

$$\frac{\partial g}{\partial I} = \beta S - \gamma$$

Step 5. Build J:

$$J = \begin{pmatrix} -\beta I & -\beta S \\ \beta I & \beta S - \gamma \end{pmatrix}$$

Step 6. Evaluate at $(S^*, I^*) = (N, 0)$. Plug in $I = 0$ and $S = N$:

$$J^* = \begin{pmatrix} -\beta \cdot 0 & -\beta \cdot N \\ \beta \cdot 0 & \beta N - \gamma \end{pmatrix} = \begin{pmatrix} 0 & -\beta N \\ 0 & \beta N - \gamma \end{pmatrix}$$

($\beta \cdot 0 = 0$)

(d) Eigenvalues:
Step 1. Notice J^* is upper triangular (the (2,1) entry is 0).
Step 2. For triangular matrices, eigenvalues are the diagonal entries:

$$\lambda_1 = 0, \qquad \lambda_2 = \beta N - \gamma$$

Step 3. Factor γ from λ_2:

$$\lambda_2 = \beta N - \gamma = \gamma \left(\frac{\beta N}{\gamma} - 1 \right) = \gamma(R_0 - 1)$$

where

$$\boxed{R_0 = \frac{\beta N}{\gamma}}$$

(e) Stability conditions:

- $R_0 > 1$: $\lambda_2 = \gamma(R_0 - 1) > 0$, disease-free FP **unstable**, disease invades.
- $R_0 < 1$: $\lambda_2 < 0$, disease-free FP **stable**, disease dies out.
- $\lambda_1 = 0$: marginal direction along the continuous family of disease-free FPs (susceptible-direction freedom).

Why the A matrix fails here: $I^* = 0$ at the disease-free FP. The interaction matrix method divides by state variables ($\dot{I}/I$), which is undefined when $I = 0$. Must use standard Jacobian.

3.2 Problem 3.2: SIRS with Loss of Immunity (2024 Q3 style)

Problem

$$\frac{dS}{dt} = -\beta SI + \nu R, \quad \frac{dI}{dt} = \beta SI - \gamma I, \quad \frac{dR}{dt} = \gamma I - \nu R$$

(a) Apply QSSA-like reasoning: which compartment to set ˙ to zero?

(b) Set $dS/dt = 0$ and solve for dI/dt as function of S.

(c) Try other QSSA choices and compare.

Solution

(a) Which compartment to zero out:
Step 1. Recall MM: $S + E \rightleftharpoons [SE] \to P + E$. We set $d[SE]/dt = 0$ because the complex is the fast intermediate.
Step 2. In SIRS, the analog of $[SE]$ (intermediate compartment) is I (between S and R). *Naively*, set $dI/dt = 0$.
Step 3. But: I is what we want to predict (analog of product P, not complex $[SE]$).
Step 4. Better choice: set $dS/dt = 0$. This treats S as quasi-conserved by quick replenishment from R (rate ν).

(b) Set $dS/dt = 0$:

Step 1. The condition gives:

$$-\beta SI + \nu R = 0$$

Step 2. Solve for R:

$$\nu R = \beta SI \quad \Longrightarrow \quad R = \frac{\beta SI}{\nu}$$

(add βSI to both sides, divide by ν)

Step 3. Substitute into conservation $N = S + I + R$:

$$N = S + I + \frac{\beta SI}{\nu}$$

Step 4. Rearrange to isolate I:

$$N - S = I + \frac{\beta SI}{\nu}$$

Step 5. Factor I from RHS:

$$N - S = I\left(1 + \frac{\beta S}{\nu}\right)$$

Step 6. Get common denominator inside parens:

$$1 + \frac{\beta S}{\nu} = \frac{\nu + \beta S}{\nu}$$

Step 7.

$$N - S = I \cdot \frac{\nu + \beta S}{\nu} \quad \Longrightarrow \quad I = \frac{(N-S)\nu}{\nu + \beta S}$$

(multiply both sides by $\nu/(\nu + \beta S)$)

Step 8. Now compute dI/dt from the original equation:

$$\frac{dI}{dt} = \beta SI - \gamma I = I(\beta S - \gamma)$$

Step 9. Substitute I from step 7:

$$\frac{dI}{dt} = \frac{(N-S)\nu}{\nu + \beta S}(\beta S - \gamma) = \frac{(N-S)\nu(\beta S - \gamma)}{\nu + \beta S}$$

Behavior of dI/dt vs S:

- $S = N$: numerator factor $(N - S) = 0$, so $\dot{I} = 0$. (Disease can't grow if everyone is susceptible at $t = 0$? No — this IC point is also a fixed point.)
- $S = \gamma/\beta$: numerator factor $(\beta S - \gamma) = 0$, so $\dot{I} = 0$ at the threshold $R_0 = 1$.
- $S \in (\gamma/\beta, N)$: both numerator factors positive, $\dot{I} > 0$, disease grows.
- $S < \gamma/\beta$: $(\beta S - \gamma) < 0$, $\dot{I} < 0$, disease shrinks.

(c) Try alternative: set $dR/dt = 0$:

Step 1. The condition gives:

$$\gamma I - \nu R = 0 \quad \Longrightarrow \quad \nu R = \gamma I \quad \Longrightarrow \quad R = \gamma I/\nu$$

Step 2. Substitute into conservation:

$$N = S + I + \frac{\gamma I}{\nu} = S + I\left(1 + \frac{\gamma}{\nu}\right)$$

Step 3. Common denominator:

$$N - S = I \cdot \frac{\nu + \gamma}{\nu}$$

Step 4. Solve for I:

$$I = \frac{(N-S)\nu}{\nu+\gamma}$$

Step 5. Substitute into $\dot{I} = I(\beta S - \gamma)$:

$$\frac{dI}{dt} = \frac{(N-S)\nu(\beta S-\gamma)}{\nu+\gamma}$$

Step 6. Same zeros (at $S = N$ and $S = \gamma/\beta$), but the denominator is $\nu + \gamma$ (a constant) instead of $\nu + \beta S$ (depends on S). So this version is **linear in S between zeros**, while the previous was **hyperbolic**.
Conclusion: Different QSSA choices give different functional forms (linear vs. hyperbolic), but the threshold conditions ($R_0 = 1$ at $S = \gamma/\beta$) are the same.

3.3 Problem 3.3: SIR with Asymptomatic and Symptomatic (2020 Q4)

Problem

Compartments: S, I_a (asymptomatic), I_s (symptomatic), R. Fraction f become asymptomatic.

$$\frac{dS}{dt} = -\beta_a I_a S - \beta_s I_s S$$

$$\frac{dI_a}{dt} = f\beta_a I_a S + f\beta_s I_s S - \gamma_a I_a$$

$$\frac{dI_s}{dt} = (1-f)\beta_a I_a S + (1-f)\beta_s I_s S - \gamma_s I_s$$

$$\frac{dR}{dt} = \gamma_a I_a + \gamma_s I_s$$

(a) Show conservation. Reduce equations.

(b) Find fixed points.

(c) Compute Jacobian at disease-free FP.

(d) Find eigenvalues for special case $f = 1$ (all asymptomatic).

(e) Same for $\beta_a = \beta_s$, $\gamma_a = \gamma_s$.

Solution

(a) Conservation:
Step 1. Sum all four equations:

$$\dot{S} + \dot{I}_a + \dot{I}_s + \dot{R} = (-\beta_a I_a S - \beta_s I_s S) + [f\beta_a I_a S + f\beta_s I_s S - \gamma_a I_a]$$
$$+[(1-f)\beta_a I_a S + (1-f)\beta_s I_s S - \gamma_s I_s] + (\gamma_a I_a + \gamma_s I_s)$$

Step 2. Group terms involving $\beta_a I_a S$:

$$-\beta_a I_a S + f\beta_a I_a S + (1-f)\beta_a I_a S = \beta_a I_a S(-1 + f + (1-f)) = \beta_a I_a S(0) = 0$$

($f + (1-f) = 1$, then $-1 + 1 = 0$)
Step 3. Similarly for $\beta_s I_s S$ terms: sum to 0.
Step 4. The $\gamma_a I_a$ and $\gamma_s I_s$ terms also cancel between $\dot{I}_a, \dot{I}_s$ and $\dot{R}$.
Step 5. So $\sum \dot{X}_i = 0$ and $N = S + I_a + I_s + R$ is conserved. Eliminate R.

(b) Disease-free FP:

$$I_a^* = I_s^* = 0, \quad S^* = N, \quad R^* = 0$$

(c) Jacobian at disease-free FP (3 variables: S, I_a, I_s):

Step 1. Compute partials. For $f = -\beta_a I_a S - \beta_s I_s S$:

$$\partial f/\partial S = -\beta_a I_a - \beta_s I_s, \quad \partial f/\partial I_a = -\beta_a S, \quad \partial f/\partial I_s = -\beta_s S$$

Step 2. For $g = f\beta_a I_a S + f\beta_s I_s S - \gamma_a I_a$:

$$\partial g/\partial S = f(\beta_a I_a + \beta_s I_s)$$

$$\partial g/\partial I_a = f\beta_a S - \gamma_a, \quad \partial g/\partial I_s = f\beta_s S$$

Step 3. For $h = (1-f)\beta_a I_a S + (1-f)\beta_s I_s S - \gamma_s I_s$:

$$\partial h/\partial S = (1-f)(\beta_a I_a + \beta_s I_s)$$

$$\partial h/\partial I_a = (1-f)\beta_a S, \quad \partial h/\partial I_s = (1-f)\beta_s S - \gamma_s$$

Step 4. Build J then evaluate at $(N, 0, 0)$. With $I_a = I_s = 0$, the first column becomes all zeros and $S = N$:

$$J^* = \begin{pmatrix} 0 & -\beta_a N & -\beta_s N \\ 0 & f\beta_a N - \gamma_a & f\beta_s N \\ 0 & (1-f)\beta_a N & (1-f)\beta_s N - \gamma_s \end{pmatrix}$$

Eigenvalues (general):
Step 1. First column zero $\Rightarrow \lambda_1 = 0$.
Step 2. Other eigenvalues come from the lower-right 2×2 block:

$$M = \begin{pmatrix} f\beta_a N - \gamma_a & f\beta_s N \\ (1-f)\beta_a N & (1-f)\beta_s N - \gamma_s \end{pmatrix}$$

Step 3. Trace:

$$\mathrm{Tr}(M) = (f\beta_a N - \gamma_a) + ((1-f)\beta_s N - \gamma_s) = f\beta_a N + (1-f)\beta_s N - \gamma_a - \gamma_s$$

Step 4. Determinant: $ad - bc$:

$$\mathrm{Det}(M) = (f\beta_a N - \gamma_a) \cdot ((1-f)\beta_s N - \gamma_s) - (f\beta_s N) \cdot ((1-f)\beta_a N)$$

Step 5. Expand the first product:

$$(f\beta_a N)((1-f)\beta_s N) + (f\beta_a N)(-\gamma_s) + (-\gamma_a)((1-f)\beta_s N) + (-\gamma_a)(-\gamma_s)$$
$$= f(1-f)\beta_a\beta_s N^2 - f\beta_a N\gamma_s - (1-f)\beta_s N\gamma_a + \gamma_a\gamma_s$$

Step 6. Subtract second product:

$$\mathrm{Det}(M) = f(1-f)\beta_a\beta_s N^2 - f\beta_a N\gamma_s - (1-f)\beta_s N\gamma_a + \gamma_a\gamma_s - f(1-f)\beta_a\beta_s N^2$$

Step 7. The $\beta_a\beta_s N^2$ terms cancel:

$$\mathrm{Det}(M) = -f\beta_a N\gamma_s - (1-f)\beta_s N\gamma_a + \gamma_a\gamma_s$$

Step 8. Factor $\gamma_a\gamma_s$:

$$\mathrm{Det}(M) = \gamma_a\gamma_s \left[1 - \frac{f\beta_a N}{\gamma_a} - \frac{(1-f)\beta_s N}{\gamma_s}\right]$$

(divide each term by $\gamma_a\gamma_s$ inside)

(d) Case $f = 1$ (all asymptomatic):
Plug $f = 1$: every coefficient with $(1-f)$ becomes 0:

$$J^* = \begin{pmatrix} 0 & -\beta_a N & -\beta_s N \\ 0 & \beta_a N - \gamma_a & \beta_s N \\ 0 & 0 & -\gamma_s \end{pmatrix}$$

Step 1. Triangular (with first column and (3,1), (3,2) zero). Eigenvalues are diagonal:

$$\lambda_1 = 0, \quad \lambda_2 = \beta_a N - \gamma_a, \quad \lambda_3 = -\gamma_s$$

Step 2. $\lambda_3 = -\gamma_s < 0$ (always). $\lambda_2 > 0$ iff $\beta_a N > \gamma_a$, i.e. $R_0 = \beta_a N/\gamma_a > 1$.
Biological meaning: With $f = 1$, all infections produce asymptomatic. The I_s compartment is unused; only the asymptomatic-disease R_0 matters.

(e) Case $\beta_a = \beta_s = \beta$, $\gamma_a = \gamma_s = \gamma$:
Step 1. Plug in:

$$M = \begin{pmatrix} f\beta N - \gamma & f\beta N \\ (1-f)\beta N & (1-f)\beta N - \gamma \end{pmatrix}$$

Step 2. Trace:

$$\mathrm{Tr}(M) = f\beta N - \gamma + (1-f)\beta N - \gamma = \beta N(f + (1-f)) - 2\gamma = \beta N - 2\gamma$$

Step 3. Determinant. From step 7 of general case with these substitutions:

$$\mathrm{Det}(M) = \gamma^2[1 - f\beta N/\gamma - (1-f)\beta N/\gamma] = \gamma^2[1 - \beta N/\gamma] = \gamma(\gamma - \beta N) = -\gamma(\beta N - \gamma)$$

Step 4. For eigenvalues, factor: $\lambda^2 - (\beta N - 2\gamma)\lambda + (-\gamma)(\beta N - \gamma) = 0$. Try $\lambda_+ = \beta N - \gamma$:

$$(\beta N - \gamma)^2 - (\beta N - 2\gamma)(\beta N - \gamma) + (-\gamma)(\beta N - \gamma)$$

Factor $(\beta N - \gamma)$:

$$= (\beta N - \gamma)[(\beta N - \gamma) - (\beta N - 2\gamma) - \gamma] = (\beta N - \gamma)[\beta N - \gamma - \beta N + 2\gamma - \gamma] = (\beta N - \gamma) \cdot 0 = 0 \checkmark$$

Step 5. So $\lambda_+ = \beta N - \gamma$. Then $\lambda_- = \mathrm{Tr}(M) - \lambda_+ = (\beta N - 2\gamma) - (\beta N - \gamma) = -\gamma$.
Step 6. Combined eigenvalues of J^*: $\lambda = 0$, $\beta N - \gamma$, $-\gamma$.
Step 7. Disease spreads iff $\lambda_+ = \beta N - \gamma > 0$, i.e. $R_0 = \beta N/\gamma > 1$. Same as standard SIR.
Biological meaning: When asymptomatic and symptomatic infections are identical (same β, same γ), the split into compartments I_a and I_s doesn't change the R_0. This makes sense: same biology, just labeled differently.

3.4 Problem 3.4: Two-Variant Disease (2021 Q3-Q4)

Problem

First virus then variant. Variant has half transmissibility for those who already had the first virus.

$$\frac{dS}{dt} = -\beta SI - \beta_2 SI_2$$
$$\frac{dI}{dt} = \beta SI - \gamma I$$
$$\frac{dS_2}{dt} = \gamma I - \frac{\beta_2}{2} S_2 I_2$$
$$\frac{dI_2}{dt} = \beta_2 I_2 (S + S_2/2) - \gamma I_2$$
$$\frac{dR}{dt} = \gamma I_2$$

(a) Conservation? Reduce.

(b) Find fixed points.

(c) Which FP for stability analysis of variant?

(d) Compute Jacobian at that FP.

(e) Find eigenvalues. Express conditions in terms of R_0.

Solution

(a) Sum all 5: cancels to 0. So $N = S + I + S_2 + I_2 + R$ const. Eliminate R.
(b) Fixed points: Need $\dot{I} = 0 \Rightarrow I^* = 0$ or $S^* = \gamma/\beta$. Need $\dot{I}_2 = 0 \Rightarrow I_2^* = 0$ or $\beta_2(S^* + S_2^*/2) = \gamma$. The non-trivial coexistence with both viruses circulating doesn't have a clean fixed point in general — must explore special cases:

- All recovered: $S = I = S_2 = I_2 = 0$, $R = N$.
- Pre-outbreak: $S = N$, all others 0.
- **First virus done, variant not yet**: $I^* = I_2^* = 0$, $S^* = N - S_2^*$ (anyone left over) or S_2^* chosen freely. This is the relevant FP.

(c) To assess if variant takes off: perturb around the post-first-virus state, $I^* = I_2^* = R^* = 0$, $S_2^* =$ whatever fraction was infected by first virus, $S^* = N - S_2^*$. Use **standard Jacobian** (compartments are zero, so A-matrix approach fails).
(d) Jacobian at $(S, I, S_2, I_2) = (N - S_2^*, 0, S_2^*, 0)$**:** $\partial(\dot{S})/\partial S = -\beta I - \beta_2 I_2 \to 0$
$\partial(\dot{S})/\partial I = -\beta S \to -\beta(N - S_2^*)$
$\partial(\dot{S})/\partial S_2 = 0$
$\partial(\dot{S})/\partial I_2 = -\beta_2 S \to -\beta_2(N - S_2^*)$
$\partial(\dot{I})/\partial I = \beta S - \gamma \to \beta(N - S_2^*) - \gamma$
$\partial(\dot{S}_2)/\partial I = \gamma$, $\partial(\dot{S}_2)/\partial I_2 = -\beta_2 S_2^*/2$
$\partial(\dot{I}_2)/\partial I_2 = \beta_2(S + S_2/2) - \gamma \to \beta_2(N - S_2^*/2) - \gamma$

$$J^* = \begin{pmatrix} 0 & -\beta(N - S_2^*) & 0 & -\beta_2(N - S_2^*) \\ 0 & \beta(N - S_2^*) - \gamma & 0 & 0 \\ 0 & \gamma & 0 & -\beta_2 S_2^*/2 \\ 0 & 0 & 0 & \beta_2(N - S_2^*/2) - \gamma \end{pmatrix}$$

(e) Eigenvalues: Block-triangular with two zero columns (S and S_2). $\lambda_1 = \lambda_2 = 0$ (susceptible directions; these are redundant constraint directions). $\lambda_3 = \beta(N - S_2^*) - \gamma$ (first virus mode). $\lambda_4 = \beta_2(N - S_2^*/2) - \gamma$ (variant mode).
Conditions:

- First virus spreads if $\lambda_3 > 0 \iff \beta(N - S_2^*)/\gamma > 1 \iff R_{0,1} > 1$.
- Variant spreads if $\lambda_4 > 0 \iff \beta_2(N - S_2^*/2)/\gamma > 1 \iff R_{0,2} > 1$ where $R_{0,2} = \beta_2(N - S_2^*/2)/\gamma$.

Two separate R_0 values make sense: each strain has its own reproductive number conditional on the susceptible pool (which differs since variant gets reduced spread from S_2 population).

3.5 Problem 3.5: Vaccination (2021 Q4d-e)

Problem

Fraction f vaccinated. Vaccine reduces infection rate $\beta_2 \to \beta_2(1-p_1)$ and increases recovery $\gamma \to \gamma(1+p_2)$ for the vaccinated. Find: how do infection terms change? What's condition for variant to spread when $f=1$?

Solution

For each contact event, both partners might be vaccinated (probability f^2), one of two scenarios for one being vaccinated ($2f(1-f)$), or neither ($(1-f)^2$). Effective infection rate (averaged over pairings):

$$\beta_2^{\text{eff}} = \beta_2[(1-f)^2 + 2f(1-f)(1-p_1) + f^2(1-p_1)^2] = \beta_2[1-f(1-(1-p_1))]^2 = \beta_2(1-fp_1)^2$$

Wait, more carefully — for a transmission to happen, only one direction matters typically. Common approximation: factor of $(1-fp_1)$ once for transmitter (reduced shedding) and once for receiver (immunity). So:

$$\beta_2 \to \beta_2(1-fp_1)^2$$

Similarly recovery for vaccinated: $\gamma \to \gamma(1+fp_2)$.
When $f=1$:

$$\beta_2^{\text{eff}} = \beta_2(1-p_1)^2, \quad \gamma^{\text{eff}} = \gamma(1+p_2)$$

Variant spreads if:

$$\beta_2(1-p_1)^2(N - S_2^*/2) > \gamma(1+p_2)$$

$\iff (1-p_1)^2 R_{0,2} > 1+p_2$
$\iff R_{0,2}^{\text{vacc}} = \frac{(1-p_1)^2 R_{0,2}}{1+p_2} > 1.$
Larger p_1 (better immunity) and larger p_2 (faster recovery) reduce variant spread. Vaccines help.

3.6 Problem 3.6: SEIR Model (Exposed Compartment)

Problem

Add an **exposed** (latent) compartment E to SIR. Infected individuals first pass through an exposed/incubating phase before becoming infectious:

$$\frac{dS}{dt} = -\beta SI, \quad \frac{dE}{dt} = \beta SI - \sigma E, \quad \frac{dI}{dt} = \sigma E - \gamma I, \quad \frac{dR}{dt} = \gamma I$$

where σ is the rate of progressing from exposed to infectious (incubation rate).

(a) Verify conservation and reduce.

(b) Find the disease-free FP.

(c) Compute J^* at the disease-free FP.

(d) Find eigenvalues and R_0.

Solution

(a) Conservation:
Step 1. Sum all four equations:

$$\dot{S} + \dot{E} + \dot{I} + \dot{R} = (-\beta SI) + (\beta SI - \sigma E) + (\sigma E - \gamma I) + \gamma I$$

Step 2. Cancel pairs: $-\beta SI + \beta SI = 0$, $-\sigma E + \sigma E = 0$, $-\gamma I + \gamma I = 0$:

$$= 0 \checkmark$$

$N = S + E + I + R$ is conserved. Eliminate R; work with (S, E, I).

(b) Disease-free FP:
Step 1. $E^* = I^* = 0$, $S^* = N$, $R^* = 0$.

(c) Jacobian at $(N, 0, 0)$:
Step 1. Define $f = -\beta SI$, $g = \beta SI - \sigma E$, $h = \sigma E - \gamma I$. Compute partials.
Step 2. Partials of f:

$$\partial f/\partial S = -\beta I, \quad \partial f/\partial E = 0, \quad \partial f/\partial I = -\beta S$$

Step 3. Partials of g:

$$\partial g/\partial S = \beta I, \quad \partial g/\partial E = -\sigma, \quad \partial g/\partial I = \beta S$$

Step 4. Partials of h:

$$\partial h/\partial S = 0, \quad \partial h/\partial E = \sigma, \quad \partial h/\partial I = -\gamma$$

Step 5. Evaluate at $(S, E, I) = (N, 0, 0)$:

$$J^* = \begin{pmatrix} 0 & 0 & -\beta N \\ 0 & -\sigma & \beta N \\ 0 & \sigma & -\gamma \end{pmatrix}$$

(Setting $I = 0$ zeros out all terms involving βI; substituting $S = N$)

(d) Eigenvalues:
Step 1. First column is all zeros: $\lambda_1 = 0$.
Step 2. Remaining 2×2 sub-matrix (from E, I rows):

$$M = \begin{pmatrix} -\sigma & \beta N \\ \sigma & -\gamma \end{pmatrix}$$

Step 3. Trace:

$$\mathrm{Tr}(M) = -\sigma - \gamma < 0$$

Step 4. Determinant:

$$\mathrm{Det}(M) = (-\sigma)(-\gamma) - (\beta N)(\sigma) = \sigma\gamma - \beta N\sigma = \sigma(\gamma - \beta N)$$

Step 5. For disease-free FP to be stable, need Det > 0:

$$\sigma(\gamma - \beta N) > 0 \quad \Longrightarrow \quad \gamma > \beta N \quad \Longrightarrow \quad \frac{\beta N}{\gamma} < 1 \quad \Longrightarrow \quad R_0 < 1$$

($\sigma > 0$ always, so sign of Det is determined by $\gamma - \beta N$)
Step 6. Define:

$$\boxed{R_0 = \frac{\beta N}{\gamma}}$$

Same formula as standard SIR! The exposed compartment adds a delay but does not change R_0.
Step 7. Eigenvalues via quadratic formula:

$$\lambda_\pm = \frac{-(\sigma + \gamma) \pm \sqrt{(\sigma + \gamma)^2 - 4\sigma(\gamma - \beta N)}}{2}$$

Step 8. Disease spreads iff Det < 0, i.e. $R_0 > 1$. When $R_0 > 1$: Det < 0, discriminant $> (\sigma + \gamma)^2$, so $\sqrt{\ } > \sigma + \gamma$, making $\lambda_+ > 0$. Disease invades.

3.7 Problem 3.7: SIR with Births and Deaths

Problem

Add equal birth and death rates μ to the standard SIR:

$$\frac{dS}{dt} = \mu N - \beta SI - \mu S$$

$$\frac{dI}{dt} = \beta SI - \gamma I - \mu I$$

$$\frac{dR}{dt} = \gamma I - \mu R$$

where the μN term is births (all born susceptible) and $-\mu X$ terms are deaths in each compartment.

(a) Verify conservation (or lack thereof). Does N change?

(b) Find the disease-free FP.

(c) Compute J^*, find eigenvalues, define R_0.

(d) How does births/deaths change R_0 compared to standard SIR?

Solution

(a) Conservation check:
Step 1. Sum all three:

$$\dot{S} + \dot{I} + \dot{R} = (\mu N - \beta SI - \mu S) + (\beta SI - \gamma I - \mu I) + (\gamma I - \mu R)$$

Step 2. Collect:

$$= \mu N - \mu(S + I + R) - \beta SI + \beta SI - \gamma I + \gamma I$$

Step 3. Cancel pairs and use $S + I + R = N$:

$$= \mu N - \mu N = 0 \checkmark$$

Step 4. Total population N is **still conserved** (births balance deaths). Eliminate R.

(b) Disease-free FP:
At $I^* = 0$: $\dot{S} = \mu N - \mu S^* = 0 \Rightarrow S^* = N$. So $(S^*, I^*) = (N, 0)$.

(c) Jacobian:
Step 1. Partials of $f = \mu N - \beta SI - \mu S$:

$$\partial f/\partial S = -\beta I - \mu, \quad \partial f/\partial I = -\beta S$$

Step 2. Partials of $g = \beta SI - \gamma I - \mu I = \beta SI - (\gamma + \mu)I$:

$$\partial g/\partial S = \beta I, \quad \partial g/\partial I = \beta S - (\gamma + \mu)$$

Step 3. Build and evaluate at $(N, 0)$:

$$J^* = \begin{pmatrix} -\mu & -\beta N \\ 0 & \beta N - (\gamma + \mu) \end{pmatrix}$$

(Setting $I = 0$: $\partial f/\partial S = -\mu$; $\partial g/\partial S = 0$)
Step 4. Upper triangular $\Rightarrow$ eigenvalues are diagonal:

$$\lambda_1 = -\mu < 0, \qquad \lambda_2 = \beta N - (\gamma + \mu)$$

Step 5. Factor λ_2:

$$\lambda_2 = (\gamma + \mu)\left(\frac{\beta N}{\gamma + \mu} - 1\right) = (\gamma + \mu)(R_0 - 1)$$

where

$$\boxed{R_0 = \frac{\beta N}{\gamma + \mu}}$$

(d) Effect of births/deaths:
Standard SIR: $R_0 = \beta N/\gamma$.
SIR with births/deaths: $R_0 = \beta N/(\gamma + \mu)$.
Step 1. Since $\gamma + \mu > \gamma$, the new $R_0 <$ standard R_0.
Step 2. Births/deaths make disease *harder* to spread (smaller R_0). Why?
Step 3. The additional μ in the denominator reflects that infected individuals also die at rate μ before recovering. They are removed from the infectious pool both by recovery (γ) and by death (μ). Effectively, the **average infectious period** is $1/(\gamma + \mu)$ rather than $1/\gamma$.
Step 4. Herd immunity threshold: $f^* = 1 - 1/R_0 = 1 - (\gamma + \mu)/(\beta N)$. Slightly lower than without deaths, meaning fewer people need to be vaccinated.

4. Michaelis-Menten Kinetics

4.1 Problem 4.1: Standard QSSA Derivation

Problem

$$S + E \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} [SE] \xrightarrow{k_2} P + E$$

Total enzyme $E_0 = E + [SE]$ conserved.

(a) Write all four equations.

(b) Apply QSSA. Solve for $[SE]$ in terms of S and E_0.

(c) Find $V_{\max}$, K_M.

(d) Show limits: $S \ll K_M$ and $S \gg K_M$.

Solution

(a) Equations:
The reactions:

- $S + E \to [SE]$ at rate $k_1 SE$ (consumes S, E; produces $[SE]$)
- $[SE] \to S + E$ at rate $k_{-1}[SE]$ (reverse: consumes $[SE]$; produces S, E)
- $[SE] \to P + E$ at rate $k_2[SE]$ (consumes $[SE]$; produces P, E)

Step 1. Build each ODE by listing all rates that produce or consume that species:

$$\dot{S} = -k_1 SE + k_{-1}[SE]$$

(S consumed at rate $k_1 SE$; replenished at rate $k_{-1}[SE]$)

$$\dot{E} = -k_1 SE + k_{-1}[SE] + k_2[SE] = -k_1 SE + (k_{-1} + k_2)[SE]$$

(E consumed forming complex; replenished by both back-reactions; combine like terms)

$$\dot{[SE]} = k_1 SE - k_{-1}[SE] - k_2[SE] = k_1 SE - (k_{-1} + k_2)[SE]$$

$$\dot{P} = k_2[SE]$$

Step 2. Verify conservation. Sum $\dot{E} + \dot{[SE]}$:

$$[-k_1 SE + (k_{-1} + k_2)[SE]] + [k_1 SE - (k_{-1} + k_2)[SE]]$$

$$= -k_1 SE + k_1 SE + (k_{-1} + k_2)[SE] - (k_{-1} + k_2)[SE] = 0 \checkmark$$

So $E + [SE] = E_0$ constant.

(b) QSSA derivation:
Step 1. Set $d[SE]/dt = 0$:

$$k_1 SE - (k_{-1} + k_2)[SE] = 0$$

Step 2. Move the second term to RHS:

$$k_1 SE = (k_{-1} + k_2)[SE]$$

Step 3. Use conservation $E = E_0 - [SE]$. Substitute:

$$k_1 S(E_0 - [SE]) = (k_{-1} + k_2)[SE]$$

Step 4. Distribute $k_1 S$ on LHS:

$$k_1 SE_0 - k_1 S[SE] = (k_{-1} + k_2)[SE]$$

Step 5. Move $-k_1S[SE]$ to RHS:

$$k_1SE_0 = (k_{-1} + k_2)[SE] + k_1S[SE]$$

Step 6. Factor $[SE]$ on RHS:

$$k_1SE_0 = [SE] \cdot (k_{-1} + k_2 + k_1S)$$

Step 7. Divide both sides by $(k_{-1} + k_2 + k_1S)$:

$$[SE] = \frac{k_1SE_0}{k_{-1} + k_2 + k_1S}$$

Step 8. Divide numerator and denominator by k_1:

$$[SE] = \frac{SE_0}{(k_{-1} + k_2)/k_1 + S}$$

($k_1S/k_1 = S$ in numerator; $k_1S/k_1 = S$ in denominator; $(k_{-1} + k_2)/k_1$ remains)

Step 9. Define $K_M = (k_{-1} + k_2)/k_1$:

$$\boxed{[SE] = \frac{E_0S}{K_M + S}}$$

(c) Production rate:

Step 1. From $\dot{P} = k_2[SE]$, substitute the QSSA result:

$$\frac{dP}{dt} = k_2 \cdot \frac{E_0S}{K_M + S} = \frac{k_2E_0S}{K_M + S}$$

Step 2. Define $V_{\max} = k_2E_0$:

$$\boxed{\frac{dP}{dt} = \frac{V_{\max}S}{K_M + S}}$$

(d) Limits:

Limit 1: $S \ll K_M$**.** Denominator $K_M + S \approx K_M$:

$$\frac{dP}{dt} \approx \frac{V_{\max}S}{K_M} = \frac{V_{\max}}{K_M} \cdot S$$

Linear in S (first-order kinetics). Slope is $V_{\max}/K_M$.

Limit 2: $S \gg K_M$**.** Denominator $K_M + S \approx S$:

$$\frac{dP}{dt} \approx \frac{V_{\max}S}{S} = V_{\max}$$

Constant (zero-order kinetics; saturation).

At $S = K_M$**:**

$$\frac{dP}{dt} = \frac{V_{\max} \cdot K_M}{K_M + K_M} = \frac{V_{\max}K_M}{2K_M} = \frac{V_{\max}}{2}$$

Half-maximum velocity (defining property).

4.2 Problem 4.2: Spike Protein Coupled Reaction (2023 Q4)

Problem

$$S + E \rightleftharpoons [SE] \rightarrow P + E$$

$$S + \tilde{\sigma} \rightleftharpoons [S\tilde{\sigma}] \rightarrow P_{\text{new}} + \tilde{\sigma}$$

with rate constants k_1, k_{-1}, k_2 for the first reaction and $\tilde{k}_1, \tilde{k}_{-1}, \tilde{k}_2$ for the second. Spike protein $\tilde{\sigma}$ acts like a second enzyme.

(a) Write all equations.

(b) How many independent equations after conservation?

(c) Apply QSSA. Find $\dot{P}$ and $\dot{P}_{\text{new}}$.

(d) How would degradation modify this?

Solution

(a) All ODEs:
The reactions involve S binding both E (rate k_1) and $\tilde{\sigma}$ (rate $\tilde{k}_1$), with both reactions producing products. Build each ODE by listing rates that consume or produce that species:

$$\dot{S} = -k_1 SE + k_{-1}[SE] - \tilde{k}_1 S\tilde{\sigma} + \tilde{k}_{-1}[S\tilde{\sigma}]$$

(S consumed forming both complexes; replenished by both back-reactions)

$$\dot{E} = -k_1 SE + (k_{-1} + k_2)[SE]$$

$$\dot{[SE]} = k_1 SE - (k_{-1} + k_2)[SE]$$

$$\dot{\tilde{\sigma}} = -\tilde{k}_1 S\tilde{\sigma} + (\tilde{k}_{-1} + \tilde{k}_2)[S\tilde{\sigma}]$$

$$\dot{[S\tilde{\sigma}]} = \tilde{k}_1 S\tilde{\sigma} - (\tilde{k}_{-1} + \tilde{k}_2)[S\tilde{\sigma}]$$

$$\dot{P} = k_2[SE], \qquad \dot{P}_{\text{new}} = \tilde{k}_2[S\tilde{\sigma}]$$

That's 7 equations.

(b) Conservation laws:
Step 1. Verify first conservation: $\dot{E} + \dot{[SE]}$:

$$[-k_1 SE + (k_{-1} + k_2)[SE]] + [k_1 SE - (k_{-1} + k_2)[SE]] = 0\checkmark$$

So $E + [SE] = E_0$.
Step 2. Verify second: $\dot{\tilde{\sigma}} + \dot{[S\tilde{\sigma}]}$:

$$[-\tilde{k}_1 S\tilde{\sigma} + (\tilde{k}_{-1} + \tilde{k}_2)[S\tilde{\sigma}]] + [\tilde{k}_1 S\tilde{\sigma} - (\tilde{k}_{-1} + \tilde{k}_2)[S\tilde{\sigma}]] = 0\checkmark$$

So $\tilde{\sigma} + [S\tilde{\sigma}] = \tilde{\sigma}_0$.
Step 3. Two conservation laws $\Rightarrow$ eliminate E and $\tilde{\sigma}$. Independent equations: $7 - 2 = 5$.

(c) QSSA:
Step 1. Apply QSSA to first complex: $\dot{[SE]} = 0$:

$$k_1 SE = (k_{-1} + k_2)[SE]$$

Step 2. Substitute $E = E_0 - [SE]$:

$$k_1 S(E_0 - [SE]) = (k_{-1} + k_2)[SE]$$

Step 3. Distribute and rearrange (same algebra as Problem 4.1):

$$k_1 SE_0 = [SE](k_{-1} + k_2 + k_1 S)$$

Step 4. Solve, divide by k_1:

$$[SE] = \frac{SE_0}{K_M + S}, \quad K_M = \frac{k_{-1} + k_2}{k_1}$$

Step 5. Same procedure for second complex $[\dot{S\tilde{\sigma}}] = 0$:

$$\tilde{k}_1 S\tilde{\sigma} = (\tilde{k}_{-1} + \tilde{k}_2)[S\tilde{\sigma}]$$

Step 6. Substitute $\tilde{\sigma} = \tilde{\sigma}_0 - [S\tilde{\sigma}]$:

$$\tilde{k}_1 S(\tilde{\sigma}_0 - [S\tilde{\sigma}]) = (\tilde{k}_{-1} + \tilde{k}_2)[S\tilde{\sigma}]$$

Step 7. Solve, divide by $\tilde{k}_1$:

$$[S\tilde{\sigma}] = \frac{S\tilde{\sigma}_0}{\tilde{K}_M + S}, \quad \tilde{K}_M = \frac{\tilde{k}_{-1} + \tilde{k}_2}{\tilde{k}_1}$$

Step 8. Production rates:

$$\frac{dP}{dt} = k_2[SE] = \frac{k_2 E_0 S}{K_M + S}$$

$$\frac{dP_{\text{new}}}{dt} = \tilde{k}_2[S\tilde{\sigma}] = \frac{\tilde{k}_2\tilde{\sigma}_0 S}{\tilde{K}_M + S}$$

Important observation: The two reactions *compete* for the substrate S (it's consumed by both pathways). But because E_0 and $\tilde{\sigma}_0$ are separately conserved, in QSSA each reaction's rate depends only on S via its own MM expression. The competition shows up in how S depletes over time, not in either rate directly.

(d) With degradation:
Step 1. Modify equations:

$$\dot{P} = k_2[SE] - k_d P$$

Step 2. Setting $\dot{P} = 0$ (steady state):

$$k_2[SE] = k_d P^* \quad \Longrightarrow \quad P^* = \frac{k_2[SE]}{k_d}$$

Step 3. Substitute QSSA result for $[SE]$:

$$P^* = \frac{k_2 E_0 S}{k_d(K_M + S)}$$

Step 4. Now **both** P and P_{new} reach steady states (rather than growing forever).
Step 5. Short-term behavior: P_{new} accumulates linearly at rate $\tilde{k}_2\tilde{\sigma}_0 S/(\tilde{K}_M + S)$.
Step 6. Long-term behavior: P_{new} saturates at $P^*_{\text{new}} = \tilde{k}_2\tilde{\sigma}_0 S/[k_d(\tilde{K}_M + S)]$.
Vaccine implications: A potentially harmful product accumulates short-term but resolves long-term if degradation operates. If the byproduct triggers autoimmune feedback (additional rate boost), the system might not stabilize.

4.3 Problem 4.3: Hill Function and Cooperativity

Problem

For a reaction with cooperativity n:

$$\dot{P} = \frac{V_{\max} S^n}{K^n + S^n}$$

(a) At what S is the rate half-maximum?

(b) Compute $\frac{d(\dot{P})}{dS}$ at $S = K$. How does this depend on n?

(c) Plot the difference between $n = 1$ and $n = 4$.

Solution

(a) Half-max concentration:

Step 1. Set $\dot{P} = V_{\max}/2$:

$$\frac{V_{\max} S^n}{K^n + S^n} = \frac{V_{\max}}{2}$$

Step 2. Cancel $V_{\max}$ from both sides:

$$\frac{S^n}{K^n + S^n} = \frac{1}{2}$$

Step 3. Cross-multiply:

$$2S^n = K^n + S^n$$

Step 4. Subtract S^n from both sides:

$$S^n = K^n$$

Step 5. Take n-th root:

$$S = K$$

The half-maximum concentration is $S = K$ **regardless of** n.

(b) Slope at half-max:

Step 1. Compute $\frac{d(\dot{P})}{dS}$ using the quotient rule. Let $u = V_{\max} S^n$ and $v = K^n + S^n$. Then $u' = V_{\max} \cdot nS^{n-1}$ and $v' = nS^{n-1}$:

$$\frac{d(\dot{P})}{dS} = \frac{u'v - uv'}{v^2}$$

Step 2. Numerator:

$$u'v - uv' = V_{\max} nS^{n-1}(K^n + S^n) - V_{\max} S^n \cdot nS^{n-1}$$

Step 3. Factor $V_{\max} nS^{n-1}$ out:

$$= V_{\max} nS^{n-1}[(K^n + S^n) - S^n] = V_{\max} nS^{n-1} \cdot K^n$$

$((K^n + S^n) - S^n = K^n)$

Step 4. So:

$$\frac{d(\dot{P})}{dS} = \frac{V_{\max} nK^n S^{n-1}}{(K^n + S^n)^2}$$

Step 5. Evaluate at $S = K$. First, $S^{n-1} = K^{n-1}$:

$$\left.\frac{d(\dot{P})}{dS}\right|_{S=K} = \frac{V_{\max} nK^n \cdot K^{n-1}}{(K^n + K^n)^2}$$

Step 6. Simplify numerator: $K^n \cdot K^{n-1} = K^{2n-1}$:

$$= \frac{V_{\max} nK^{2n-1}}{(2K^n)^2}$$

Step 7. Simplify denominator: $(2K^n)^2 = 4K^{2n}$:

$$= \frac{V_{\max} nK^{2n-1}}{4K^{2n}}$$

Step 8. Cancel $K^{2n-1}/K^{2n} = 1/K$:

$$= \frac{V_{\max} n}{4K}$$

$$\boxed{\left.\frac{d(\dot{P})}{dS}\right|_{S=K} = \frac{nV_{\max}}{4K}}$$

Slope at half-max scales linearly with n. Higher n means steeper switch.

(c) Comparison of $n = 1$ vs $n = 4$:

- $n = 1$ **(standard MM):** smooth hyperbolic increase. No inflection point.
- $n = 4$ **(high cooperativity):** switch-like sigmoid. Very flat at low S, sharp rise near $S = K$, then flat at high S. Inflection point at $S = K$.

Biologically, cooperative binding (high n) produces **"digital" threshold response**: gene either off or on, channel either closed or open, near a critical concentration. This is the basis of switching in many cell-signaling pathways.

5. Hodgkin-Huxley and Action Potentials

5.1 Problem 5.1: Conceptual Questions on HH

Problem

(a) Write the HH equation for membrane voltage.

(b) Explain each term physically.

(c) Why m^3h for sodium and n^4 for potassium?

(d) Sketch dynamics of V, m, h, n during an action potential.

Solution

(a)

$$C_m \frac{dV}{dt} = -g_{\text{Na}} m^3 h (V - V_{\text{Na}}) - g_K n^4 (V - V_K) - g_L (V - V_L) + I_{\text{ext}}$$

(b) Terms:

- $C_m \dot{V}$: capacitive current (charging/discharging the membrane).
- $g_{\text{Na}} m^3 h$: maximum Na conductance times fraction of channels open (m^3h).
- $(V - V_{\text{Na}})$: driving force; current = conductance × driving force.
- $g_L(V - V_L)$: passive "leak" through always-open channels.
- I_{ext}: applied stimulus.

(c) Powers:

- Sodium has 3 activation gates (each opens with prob m), and 1 inactivation gate (closes with prob $1-h$, so open with h). Net open: m^3h.
- Potassium has 4 activation gates: n^4.
- These powers come from the structural biology of the channels (subunits).

(d) Action potential phases:

1. **Resting** ($V \approx -70$ mV): $m \approx 0$, $h \approx 1$, $n \approx 0$.
2. **Depolarization**: stimulus $\to V \uparrow \to m \uparrow$ rapidly $\to$ Na current floods in $\to V \uparrow$ further (positive feedback).
3. **Peak** ($V \approx +30$ mV): m saturates at 1.
4. **Repolarization**: $h \downarrow$ slowly (Na inactivates) and $n \uparrow$ slowly (K opens). Na current shuts off; K current flows out.
5. **Hyperpolarization (afterpolarization)**: K current overshoots; V briefly below resting.
6. **Recovery**: h slowly returns to 1; n to 0; refractory period.

5.2 Problem 5.2: Gating Variable Dynamics

Problem

A gating variable x obeys:

$$\frac{dx}{dt} = \alpha_x(V)(1 - x) - \beta_x(V)x$$

(a) Find steady-state $x_\infty(V)$ and time constant $\tau_x(V)$.

(b) Rewrite the ODE in standard relaxation form.

(c) Solve for $x(t)$ if V is held constant after $t = 0$.

Solution

(a) Steady-state and time constant:
Step 1. At steady state, $\dot{x} = 0$:

$$\alpha_x(1 - x_\infty) - \beta_x x_\infty = 0$$

Step 2. Distribute α_x:

$$\alpha_x - \alpha_x x_\infty - \beta_x x_\infty = 0$$

Step 3. Group x_∞ terms:

$$\alpha_x = \alpha_x x_\infty + \beta_x x_\infty = x_\infty(\alpha_x + \beta_x)$$

Step 4. Solve for x_∞:

$$\boxed{x_\infty(V) = \frac{\alpha_x(V)}{\alpha_x(V) + \beta_x(V)}}$$

Step 5. For τ_x, expand the original ODE:

$$\dot{x} = \alpha_x(1 - x) - \beta_x x = \alpha_x - \alpha_x x - \beta_x x = \alpha_x - (\alpha_x + \beta_x)x$$

(distribute, then factor $-x$ from last two terms)
Step 6. Compare with the standard relaxation form $\dot{x} = (x_\infty - x)/\tau_x$. Reading off:

$$\boxed{\tau_x(V) = \frac{1}{\alpha_x(V) + \beta_x(V)}}$$

(b) Standard relaxation form:
Step 1. From step 5: $\dot{x} = \alpha_x - (\alpha_x + \beta_x)x$. Factor $(\alpha_x + \beta_x)$:

$$\dot{x} = (\alpha_x + \beta_x)\left[\frac{\alpha_x}{\alpha_x + \beta_x} - x\right]$$

Step 2. Recognize $\alpha_x/(\alpha_x + \beta_x) = x_\infty$ and $(\alpha_x + \beta_x) = 1/\tau_x$:

$$\boxed{\frac{dx}{dt} = \frac{x_\infty - x}{\tau_x}}$$

(c) Solution for constant V:
Step 1. If V is constant, α_x, β_x, x_∞, τ_x are all constants. Define $y = x - x_\infty$, so $\dot{y} = \dot{x}$:

$$\dot{y} = \frac{x_\infty - (y + x_\infty)}{\tau_x} = \frac{-y}{\tau_x}$$

Step 2. This is exponential decay:

$$\frac{dy}{y} = -\frac{dt}{\tau_x} \quad \Longrightarrow \quad \int \frac{dy}{y} = -\int \frac{dt}{\tau_x} \quad \Longrightarrow \quad \ln|y| = -\frac{t}{\tau_x} + C$$

Step 3. Exponentiate:

$$y(t) = y(0)\, e^{-t/\tau_x}$$

($y(0) = e^C$)
Step 4. Substitute back $y = x - x_\infty$:

$$x(t) - x_\infty = (x(0) - x_\infty)\, e^{-t/\tau_x}$$

Step 5.

$$\boxed{x(t) = x_\infty + (x(0) - x_\infty)\, e^{-t/\tau_x}}$$

Exponential approach to x_∞ with characteristic time τ_x. As $t \to \infty$, $e^{-t/\tau_x} \to 0$, so $x \to x_\infty$. ✓

6. Stability & Eigenvalue Analysis (General)

6.1 Problem 6.1: 2×2 Matrix Eigenvalue Practice

Problem

For $M = \begin{pmatrix} 2 & -1 \\ 4 & -3 \end{pmatrix}$, compute Tr, Det, and eigenvalues. Classify the fixed point.

Solution

Step 1. Identify entries: $a = 2$, $b = -1$, $c = 4$, $d = -3$.
Step 2. Trace = sum of diagonal:

$$\mathrm{Tr}(M) = a + d = 2 + (-3) = 2 - 3 = -1$$

Step 3. Determinant = $ad - bc$:

$$\mathrm{Det}(M) = (2)(-3) - (-1)(4) = -6 - (-4) = -6 + 4 = -2$$

$(-(-4) = +4)$
Step 4. Compute discriminant:

$$\Delta = \mathrm{Tr}^2 - 4\mathrm{Det} = (-1)^2 - 4(-2) = 1 - (-8) = 1 + 8 = 9$$

Step 5. Apply quadratic formula:

$$\lambda_\pm = \frac{\mathrm{Tr} \pm \sqrt{\Delta}}{2} = \frac{-1 \pm \sqrt{9}}{2} = \frac{-1 \pm 3}{2}$$

Step 6. Compute both:

$$\lambda_+ = \frac{-1+3}{2} = \frac{2}{2} = 1$$
$$\lambda_- = \frac{-1-3}{2} = \frac{-4}{2} = -2$$

Step 7. Verify with characteristic polynomial $\lambda^2 - \mathrm{Tr}\lambda + \mathrm{Det} = 0$:

$$\lambda^2 + \lambda - 2 = 0$$

Factor: $(\lambda + 2)(\lambda - 1) = 0$, giving $\lambda = -2$ or $\lambda = 1$. ✓
Step 8. Classification: One positive ($\lambda_+ = 1$), one negative ($\lambda_- = -2$). This is a **saddle point** (always unstable).
Det < 0 on its own would have told us it's a saddle (the row in the stability table for "any Tr, Det < 0").

6.2 Problem 6.2: Asymptotic Analysis of a Perturbation (2024 Q4)

Problem

A perturbation $\delta x(t)$ obeys:

$$\delta x(t) = \frac{1}{1 - \frac{1}{\lambda\tau} e^{\lambda t}/\lambda} \cdot \frac{1}{\tau}$$

(a) Behavior at early times $t \ll \tau$.

(b) Behavior at long times $t \gg \tau$.

(c) Sketch in phase space.

(d) If this is only a leading-order Taylor expansion, what's the conclusion?

(e) For a different system: $\delta x(t) = (1 + e^{-\lambda t})e^{-\lambda t}$. Long-time behavior?

Solution

(a) Early times: $e^{\lambda t}$ near 1, so denominator $\approx 1 - 1/(\lambda^2\tau) \approx 1$ if $\lambda^2\tau \gg 1$. Leading behavior $\delta x \approx 1/\tau \cdot e^{\lambda t}/\text{stuff}$. More carefully: at small t, expand exponential, $\delta x \approx \frac{1}{\tau} \cdot e^{\lambda t}$ (exponential growth).
(b) Long times: $e^{\lambda t} \to \infty$ (since $\lambda > 0$), denominator $\to -\infty$ in magnitude, so $\delta x \to 0$. Perturbation **decays back to FP eventually**.
(c) Phase space: Trajectory first moves away from FP, then turns around and comes back. Looks like a loop or excursion.
Conclusion: Since the perturbation eventually decays to zero, the system is **stable** (this is the exact solution).
(d) If only leading-order Taylor: Cannot conclude. Higher-order terms might dominate when the perturbation grows large; the linear theory only valid near FP. The trajectory might escape before higher orders bring it back, leading to true instability.
(e) Other system: $\delta x(t) = (1 + e^{-\lambda t})e^{-\lambda t} = e^{-\lambda t} + e^{-2\lambda t} \to 0$ as $t \to \infty$. Stable, with no oscillations (real exponentials).
If λ is complex, say $\lambda = a + i\omega$, then $e^{-\lambda t}$ oscillates with envelope e^{-at}. If $a > 0$, decays $\Rightarrow$ **spiral into FP**. Stable spiral (limit-cycle-like decay).

6.3 Problem 6.3: 3×3 Stability via Routh-Hurwitz

Problem

For 3×3 matrix J^*, when is the FP stable?

Solution

Characteristic polynomial: $\lambda^3 + a_2\lambda^2 + a_1\lambda + a_0 = 0$ where:

- $a_2 = -\text{Tr}(J^*)$
- $a_1 =$ sum of 2×2 principal minors
- $a_0 = -\text{Det}(J^*)$

Routh-Hurwitz conditions for all roots to have negative real parts:

1. $a_2 > 0$ (i.e. $\text{Tr}(J^*) < 0$).
2. $a_0 > 0$ (i.e. $\text{Det}(J^*) < 0$).
3. $a_2 a_1 > a_0$.

Number of invariants for $n \times n$: For $n \times n$, you need n independent equations (Tr, Det, and $n - 2$ intermediate invariants for $n > 2$). Always $n - 2$ more than 2×2 case.

6.4 Problem 6.4: Trace and Determinant Biological Meaning (2019 Q2)

Problem

For each matrix below, describe the biological meaning of Tr, Det, and predict stability.

$$A_1 = \begin{pmatrix} -r/K & -\bar{\beta} \\ \varepsilon\bar{\beta} & 0 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 0 & -\beta \\ \beta & 0 \end{pmatrix}$$

Solution

A_1 **(logistic-prey LV):**

- $\mathrm{Tr}(A_1) = -r/K$ (sum of self-interactions). Only prey self-regulates (logistic). Negative $\Rightarrow$ stabilizing.
- $\mathrm{Det}(A_1) = \varepsilon\bar{\beta}^2$ (product of cross-species interactions). Positive $\Rightarrow$ stabilizing form (no saddle).
- Eigenvalues: $\lambda_\pm = -\frac{r}{2K}\left(1 \pm \sqrt{1 - \frac{4\varepsilon\bar{\beta}^2K^2}{r^2}}\right)$. Real part negative $\Rightarrow$ **stable**.

A_2 **(basic SIR):**

- $\mathrm{Tr}(A_2) = 0$: no self-interactions among compartments.
- $\mathrm{Det}(A_2) = \beta^2 > 0$: positive cross-interactions $\Rightarrow$ oscillatory.
- $\lambda_\pm = \pm i\beta$: pure imaginary, neutral oscillations.

7. Conservation Equations & Constraint Surfaces

7.1 Problem 7.1: Identifying Conservation

Problem

For each system, determine whether $N = \sum X_i$ is conserved:

(a) SIR with deaths: $\dot{S} = -\beta SI$, $\dot{I} = \beta SI - \gamma I - \mu I$, $\dot{R} = \gamma I$.

(b) SIRS: $\dot{S} = -\beta SI + \nu R$, $\dot{I} = \beta SI - \gamma I$, $\dot{R} = \gamma I - \nu R$.

(c) Logistic predator-prey.

Solution

(a)
Step 1. Sum all three: $\dot{S} + \dot{I} + \dot{R} = (-\beta SI) + (\beta SI - \gamma I - \mu I) + (\gamma I)$.
Step 2. Cancel: $-\beta SI + \beta SI = 0$, $\gamma I - \gamma I = 0$. Remaining: $-\mu I$.
Step 3. So $\frac{d(S+I+R)}{dt} = -\mu I \leq 0$. **Not conserved**; total population decreases due to disease deaths. Cannot eliminate any equation.
(b)
Step 1. Sum: $(-\beta SI + \nu R) + (\beta SI - \gamma I) + (\gamma I - \nu R)$.
Step 2. Cancel: $-\beta SI + \beta SI = 0$, $-\gamma I + \gamma I = 0$, $\nu R - \nu R = 0$. Result: 0.
Step 3. **Conserved**. $N = S + I + R =$ const. Eliminate one equation (e.g., $R = N - S - I$).
(c)
Step 1. Sum: $(rN(1 - N/K) - \bar{\beta}NP) + (\varepsilon\bar{\beta}NP - ZP)$.
Step 2. Collect: $rN - rN^2/K - \bar{\beta}NP + \varepsilon\bar{\beta}NP - ZP = rN - rN^2/K + (\varepsilon - 1)\bar{\beta}NP - ZP$.
Step 3. This is not zero (has leftover terms). **Not conserved**. Predator-prey has no biomass conservation because $\varepsilon < 1$ means prey biomass is lost as heat, and $-ZP$ represents death.

7.2 Problem 7.2: Reducing a 5-Variable System

Problem

For the two-variant disease in Problem 3.4, show that conservation reduces 5 variables to 4 and explain why we can't reduce further despite there being multiple constraints.

Solution

Step 1. The system has S, I, S_2, I_2, R — five variables.
Step 2. Sum all five equations. Every term appears with both a positive and negative sign across the five equations (e.g., βSI appears $-$ in $\dot{S}$ and $+$ in $\dot{I}$). Net sum: 0.
Step 3. Therefore $S + I + S_2 + I_2 + R = N$ is conserved. This lets us eliminate one variable: $R = N - S - I - S_2 - I_2$. Now 4 independent equations.
Step 4. Can we reduce further? Only if there is a **second independent conservation law**. Check: is there any other linear combination $c_1S + c_2I + c_3S_2 + c_4I_2 =$ const?
Step 5. For $c_1\dot{S} + c_2\dot{I} + c_3\dot{S}_2 + c_4\dot{I}_2 = 0$ to hold identically, we need the coefficients of each product (SI, SI_2, S_2I_2) to cancel. This only works for $c_1 = c_2 = c_3 = c_4 = 1$ (total conservation). No other independent combination gives zero.
Conclusion: The system has exactly one conservation law and reduces from 5 to 4 independent equations. Cannot reduce further.

7.3 Problem 7.3: Enzyme Conservation with Inhibitor (2018 Q4 style)

Problem

Consider competitive inhibition:

$$S + E \rightleftharpoons [SE] \to P + E, \qquad I + E \rightleftharpoons [EI]$$

(a) Identify all conservation laws.

(b) Apply QSSA to both complexes.

(c) Derive the modified production rate. How does inhibitor affect $V_{\max}$ and K_M?

Solution

(a) Conservation laws:
Step 1. Sum $\dot{E} + \dot{[SE]} + \dot{[EI]}$. Using the full equations:

$$\dot{E} = -k_1 SE + (k_{-1} + k_2)[SE] - k_3 IE + k_{-3}[EI]$$

$$\dot{[SE]} = k_1 SE - (k_{-1} + k_2)[SE]$$

$$\dot{[EI]} = k_3 IE - k_{-3}[EI]$$

Sum: all terms cancel. So $E + [SE] + [EI] = E_0$ conserved.
Step 2. No conservation on S or I (they are consumed/released without matching terms).
Step 3. One conservation law: $E_0 = E + [SE] + [EI]$.

(b) QSSA on both complexes:
Step 1. Set $\dot{[SE]} = 0$:

$$k_1 SE = (k_{-1} + k_2)[SE]$$

(same as standard MM derivation)
Step 2. Set $\dot{[EI]} = 0$:

$$k_3 IE = k_{-3}[EI] \quad \Longrightarrow \quad [EI] = \frac{k_3 IE}{k_{-3}} = \frac{IE}{K_I}$$

where $K_I = k_{-3}/k_3$ is the inhibitor dissociation constant.
Step 3. From Step 1: $[SE] = k_1 SE/(k_{-1} + k_2) = SE/K_M$ where $K_M = (k_{-1} + k_2)/k_1$.
Step 4. Substitute both into conservation $E + [SE] + [EI] = E_0$:

$$E + \frac{SE}{K_M} + \frac{IE}{K_I} = E_0$$

Step 5. Factor E:

$$E\left(1 + \frac{S}{K_M} + \frac{I}{K_I}\right) = E_0 \quad \Longrightarrow \quad E = \frac{E_0}{1 + S/K_M + I/K_I}$$

Step 6. Find $[SE]$:

$$[SE] = \frac{SE}{K_M} = \frac{S}{K_M} \cdot \frac{E_0}{1 + S/K_M + I/K_I}$$

Step 7. Multiply top and bottom by K_M:

$$[SE] = \frac{SE_0}{K_M + S + SK_M/K_I \cdot \frac{1}{S} \cdot S} = \frac{SE_0}{K_M(1 + I/K_I) + S}$$

Let me redo this cleanly:

$$[SE] = \frac{SE_0/K_M}{1 + S/K_M + I/K_I}$$

Multiply numerator and denominator by K_M:

$$[SE] = \frac{SE_0}{K_M + S + K_M I/K_I} = \frac{SE_0}{K_M(1 + I/K_I) + S}$$

(c) Modified production rate:
Step 1.

$$\dot{P} = k_2[SE] = \frac{k_2 E_0 S}{K_M(1 + I/K_I) + S}$$

Step 2. Compare to standard MM $\dot{P} = V_{\max} S/(K_M + S)$:

- $V_{\max}$ is **unchanged**: $V_{\max} = k_2 E_0$.
- K_M is **increased**: effective $K_M^{\text{eff}} = K_M(1 + I/K_I)$.

Step 3. Biological meaning: the inhibitor competes with substrate S for the enzyme. At saturation ($S \gg K_M$), the reaction still reaches $V_{\max}$ (enough substrate displaces inhibitor). But at low S, the inhibitor effectively reduces affinity (larger apparent K_M means enzyme needs more S to reach half-max).

8. Mixed / Compound Problems

8.1 Problem 8.1: COVID-19 Real-World Application (2020 Q1-3)

Problem

A pathogen doubles cases every 5 days (factor of $e \approx 2.72$).

(a) Find continuous growth rate r.

(b) If 0.5% of cases are fatal with 3-week delay, find deaths $D(t)$ on day t.

(c) Tests are processed with a 3-day delay; reported cases $R(t)$?

(d) Evaluate at $t = 100$ days starting from $N(0) = 1$.

Solution

(a) $N(t) = e^{rt}$ with $N(5) = e$ requires $5r = 1 \Rightarrow r = 0.2$ /day.
(b) Cases at time $t - 21$ days (3 weeks earlier) become deaths today, with 0.5% probability:

$$D(t) = 0.005 \cdot N(t-21) = 0.005 \cdot e^{0.2(t-21)} = 0.005\, e^{-4.2}\, e^{0.2t} \approx 7.5 \times 10^{-5} e^{0.2t}$$

(c) Reported cases lag by 3 days:

$$R(t) = N(t-3) = e^{0.2(t-3)} = e^{-0.6} e^{0.2t} \approx 0.549 e^{0.2t}$$

New reported cases per day: $\Delta R(t) = R(t) - R(t-1) \approx R(t)(1 - e^{-0.2}) \approx 0.181 R(t)$.
(d) At $t = 100$:

- $N(100) = e^{20} \approx 4.85 \times 10^8$ (unrealistic; suggests population finite, model breakdown).
- $D(100) = 7.5 \times 10^{-5} \cdot e^{20} \approx 36{,}000$
- $R(100) \approx 0.549 \cdot e^{20} \approx 2.66 \times 10^8$

The numbers are unrealistic because exponential growth ignores the finite population — this is why we need SIR-type models with saturation.

8.2 Problem 8.2: Connecting SI Model to Exponential Growth (2020 Q2)

Problem

For the SI model, $\dot{I} = \beta S I - \gamma I$ with $S + I = T$ (total constant). Show how, when $I \ll T$, this reduces to exponential growth. What is the effective r?

Solution

Step 1. Use conservation: $S + I = T$ implies $S = T - I$.
Step 2. Substitute into $\dot{I}$:

$$\dot{I} = \beta(T - I)I - \gamma I$$

Step 3. Distribute $\beta(T - I)$:

$$\dot{I} = \beta T I - \beta I^2 - \gamma I$$

Step 4. Approximation: when $I \ll T$, the term βI^2 is negligible compared to $\beta T I$ (since $I^2 \ll TI$):

$$\dot{I} \approx \beta T I - \gamma I$$

Step 5. Factor I:

$$\dot{I} = I(\beta T - \gamma)$$

Step 6. Recognize this as exponential growth $\dot{I} = rI$ with:

$$\boxed{r = \beta T - \gamma}$$

Step 7. Express in terms of R_0. Factor γ:

$$r = \gamma\left(\frac{\beta T}{\gamma} - 1\right) = \gamma(R_0 - 1)$$

Step 8. Conclusions:

- $R_0 > 1 \Rightarrow r > 0 \Rightarrow$ exponential growth (early epidemic).
- $R_0 < 1 \Rightarrow r < 0 \Rightarrow$ exponential decay (disease dies out).
- $R_0 = 1 \Rightarrow r = 0 \Rightarrow$ marginal case.

Without approximation:
Step 1. Keep full equation $\dot{I} = \beta T I - \beta I^2 - \gamma I$:

$$\dot{I} = (\beta T - \gamma)I - \beta I^2$$

Step 2. Factor:

$$\dot{I} = I[(\beta T - \gamma) - \beta I]$$

Step 3. Compare to logistic form $\dot{N} = rN(1 - N/K)$ where $r = \beta T - \gamma$ and $rN/K = \beta I^2$, giving:

$$\frac{r}{K} = \beta \quad \Longrightarrow \quad K = \frac{r}{\beta} = \frac{\beta T - \gamma}{\beta}$$

Step 4. Simplify:

$$K = T - \frac{\gamma}{\beta} = T\left(1 - \frac{1}{R_0}\right)$$

(factor T, recognize $\gamma/(\beta T) = 1/R_0$)
Step 5. Conclusion: The SI model is a logistic equation with carrying capacity $K = T(1 - 1/R_0)$. As the epidemic progresses, I saturates at this fraction of the population.

8.3 Problem 8.3: Comparing Two Cities (2020 Q2b)

Problem

City A: population T_A, infection rate β_A, $R_{0,A} = 1.1$. City B: 10× population, 5× infection rate. Compare long-time infected fraction.

Solution

Step 1. Use $R_0 = \beta N/\gamma$ (where $N = T$ here) for both cities.
Step 2. City A: $R_{0,A} = \beta_A T_A/\gamma = 1.1$.
Step 3. City B parameters:

$$T_B = 10\,T_A, \quad \beta_B = 5\,\beta_A$$

Step 4. Compute $R_{0,B}$:

$$R_{0,B} = \frac{\beta_B T_B}{\gamma} = \frac{(5\beta_A)(10T_A)}{\gamma} = 50 \cdot \frac{\beta_A T_A}{\gamma}$$

Step 5. Substitute $R_{0,A} = 1.1$:

$$R_{0,B} = 50 \cdot R_{0,A} = 50 \cdot 1.1 = 55$$

Massive amplification.

Long-time infected fraction:
Step 6. For SIR, the final size equation is:

$$1 - x = e^{-R_0 x}$$

where $x =$ fraction eventually infected. Solve transcendentally.

Step 7. For City A ($R_0 = 1.1$): try $x = 0.18$:

$$1 - 0.18 = 0.82, \quad e^{-1.1 \cdot 0.18} = e^{-0.198} \approx 0.820 \checkmark$$

So $x_A \approx 0.18$ (18% infected).
Step 8. For City B ($R_0 = 55$): for any reasonable x near 1:

$$e^{-55 \cdot x} \to 0 \text{ rapidly}$$

so $1 - x_B \to 0$, meaning $x_B \to 1$ (essentially everyone infected).
Step 9. Comparison:

- Per capita: City A has 18% infected; City B has $\sim 100\%$. Roughly $5\times$ more cases per capita.
- Total cases: City A has $0.18T_A$; City B has $\sim 10T_A$ (since $T_B = 10T_A$). About $55\times$ more total cases.
- **Hospital overflow risk**: City B will face catastrophic case load.

Step 10. Lesson: R_0 is super-linear in both transmission rate and population density. Doubling either creates much greater than $2\times$ impact when it pushes R_0 above 1.

9. Model Formulation From Words (Pattern Recognition)

9.1 Problem 9.1: Translating Biology to Equations

Problem

For each scenario, write down the differential equations.

(a) Two species in symbiosis (mutualism): both grow logistically, plus each benefits proportionally from the other.

(b) A virus that mutates from strain 1 to strain 2 at rate μ, with separate SIR dynamics for each.

(c) Enzyme inhibition: an inhibitor I binds the enzyme E to form $[EI]$, reducing the active enzyme pool.

Solution

(a) Symbiosis:

$$\dot{N}_1 = r_1 N_1(1 - N_1/K_1) + \alpha_{12} N_1 N_2$$
$$\dot{N}_2 = r_2 N_2(1 - N_2/K_2) + \alpha_{21} N_1 N_2$$

(b) Mutating virus (with conversion of strain 1 to 2 in infected hosts):

$$\dot{S} = -\beta_1 S I_1 - \beta_2 S I_2$$
$$\dot{I}_1 = \beta_1 S I_1 - \gamma I_1 - \mu I_1$$
$$\dot{I}_2 = \beta_2 S I_2 - \gamma I_2 + \mu I_1$$
$$\dot{R} = \gamma(I_1 + I_2)$$

(c) Enzyme inhibition (competitive):

$$S + E \rightleftharpoons [SE] \to P + E$$
$$I + E \rightleftharpoons [EI]$$

Equations:

$$\dot{S} = -k_1 SE + k_{-1}[SE]$$
$$\dot{I} = -k_3 IE + k_{-3}[EI]$$
$$\dot{E} = -k_1 SE + (k_{-1} + k_2)[SE] - k_3 IE + k_{-3}[EI]$$
$$\dot{[SE]} = k_1 SE - (k_{-1} + k_2)[SE]$$
$$\dot{[EI]} = k_3 IE - k_{-3}[EI]$$

Conservation: $E_0 = E + [SE] + [EI]$. After QSSA on both complexes: $V_{\max}^{\text{eff}} = V_{\max}/(1 + I/K_I)$ where $K_I = k_{-3}/k_3$. Inhibitor reduces effective $V_{\max}$.

9.2 Problem 9.2: Carrying Capacity Modification (2018 Q3)

Problem

Standard LV without predator carrying capacity. What assumption underlies this? What if the assumption fails? Modify equations.

Solution

Assumption: Predator population is solely limited by prey availability (food). When prey are abundant, predators grow without saturation.
When this fails: Territoriality, intraspecific competition, mate-finding limits, or alternative resources.

Modification: Replace $\varepsilon\bar{\beta}N$ (per-capita predator growth) with $\varepsilon\bar{\beta}N(1 - P/K_P)$:

$$\frac{dP}{dt} = \varepsilon\bar{\beta}NP(1 - P/K_P) - ZP$$

Now P has its own carrying capacity. Adds negative diagonal term to A:

$$A = \begin{pmatrix} 0 & -\bar{\beta} \\ \varepsilon\bar{\beta} & -\varepsilon\bar{\beta}N^*/K_P \end{pmatrix}$$

At fixed point, this becomes stabilizing.

9.3 Problem 9.3: Chronic Carriers (2018 Q4)

Problem

SIR model with chronic carriers C:

$$\dot{S} = -\beta SI - \alpha SC + \gamma I$$
$$\dot{I} = \beta SI - \gamma I$$
$$\dot{C} = \alpha SC$$

(a) Conservation? Reduce.

(b) Find fixed points.

(c) Compute Jacobian at disease-free FP.

(d) Find eigenvalues. Define R_0 for each pathogen branch.

Solution

(a) Conservation:
Step 1. Sum all three equations:

$$\dot{S} + \dot{I} + \dot{C} = (-\beta SI - \alpha SC + \gamma I) + (\beta SI - \gamma I) + (\alpha SC)$$

Step 2. Group like terms:

$$= (-\beta SI + \beta SI) + (-\alpha SC + \alpha SC) + (\gamma I - \gamma I) = 0\checkmark$$

Step 3. So $S + I + C = N$ is conserved. Use this to eliminate $S = N - I - C$ if desired. (For 3×3 Jacobian we'll keep all three.)

(b) Fixed points:
Step 1. Set $\dot{I} = 0$:

$$\beta SI - \gamma I = I(\beta S - \gamma) = 0 \quad \Longrightarrow \quad I = 0 \text{ or } S = \gamma/\beta$$

Step 2. Set $\dot{C} = 0$:

$$\alpha SC = 0 \quad \Longrightarrow \quad S = 0 \text{ or } C = 0$$

Step 3. Disease-free: $I^* = 0$, $C^* = 0$, $S^* = N$. (Easy to verify $\dot{S} = 0$.)
Step 4. Endemic with $C^* = 0$ and $I^* > 0$: requires $S^* = \gamma/\beta$, then $I^* = N - \gamma/\beta$ (only positive if $R_0 = \beta N/\gamma > 1$).

(c) Jacobian at disease-free FP $(S, I, C) = (N, 0, 0)$:
Step 1. Define $f = \dot{S}$, $g = \dot{I}$, $h = \dot{C}$. Compute partials.
Step 2. For $f = -\beta SI - \alpha SC + \gamma I$:

$$\partial f/\partial S = -\beta I - \alpha C$$
$$\partial f/\partial I = -\beta S + \gamma$$

$$\partial f/\partial C = -\alpha S$$

Step 3. For $g = \beta SI - \gamma I$:

$$\partial g/\partial S = \beta I, \quad \partial g/\partial I = \beta S - \gamma, \quad \partial g/\partial C = 0$$

Step 4. For $h = \alpha SC$:

$$\partial h/\partial S = \alpha C, \quad \partial h/\partial I = 0, \quad \partial h/\partial C = \alpha S$$

Step 5. Build J then evaluate at $(N, 0, 0)$:

- $\partial f/\partial S \to -\beta \cdot 0 - \alpha \cdot 0 = 0$
- $\partial f/\partial I \to -\beta N + \gamma$
- $\partial f/\partial C \to -\alpha N$
- $\partial g/\partial S \to \beta \cdot 0 = 0$
- $\partial g/\partial I \to \beta N - \gamma$
- $\partial g/\partial C \to 0$
- $\partial h/\partial S \to \alpha \cdot 0 = 0$
- $\partial h/\partial I \to 0$
- $\partial h/\partial C \to \alpha N$

Step 6. Final Jacobian:

$$J^* = \begin{pmatrix} 0 & \gamma - \beta N & -\alpha N \\ 0 & \beta N - \gamma & 0 \\ 0 & 0 & \alpha N \end{pmatrix}$$

(d) Eigenvalues:
Step 1. The first column is all zeros, so $\lambda_1 = 0$.
Step 2. The remaining 2×2 sub-matrix is upper triangular:

$$M = \begin{pmatrix} \beta N - \gamma & 0 \\ 0 & \alpha N \end{pmatrix}$$

Eigenvalues are diagonal entries: $\lambda_2 = \beta N - \gamma$, $\lambda_3 = \alpha N$.
Step 3. Three eigenvalues:

$$\boxed{\lambda_1 = 0, \quad \lambda_2 = \beta N - \gamma, \quad \lambda_3 = \alpha N}$$

Step 4. Define R_0 for each transmission branch:

$$R_0^{(I)} = \frac{\beta N}{\gamma} \quad \text{(infective branch)}$$

For the chronic-carrier branch, since carriers never recover, the relevant "R_0-like" quantity is just αN (multiplied by an effective lifetime).
Step 5. Stability conditions:

- $\lambda_2 > 0 \iff R_0^{(I)} > 1 \iff$ infective branch invades.
- $\lambda_3 > 0 \iff \alpha N > 0$ **always**.

Step 6. Conclusion: **the disease-free FP is always unstable** as long as $\alpha N > 0$ (any positive transmission rate). Once a chronic carrier exists, the system cannot return to disease-free.
Why one R_0 is enough: If $\alpha > 0$, $\lambda_3 > 0$ and instability is guaranteed regardless of $R_0^{(I)}$. So for stability we'd need BOTH $\alpha = 0$ (no chronic carrier transmission) AND $R_0^{(I)} < 1$. In practice, only $R_0^{(C)}$ matters for whether the system can be stable at all.

9.4 Problem 9.4: Multiple-Pack Hunting Variant (2024 Q1f)

Problem

Now flip the pack-hunting scenario: 5 predators hunt and kill 100 prey at a time (like a dolphin pod consuming a school of fish).

(a) What are the natural compartments?

(b) Write the differential equations.

(c) Interpret parameters vs standard LV.

(d) Write A, find Tr, Det, eigenvalues, and comment on stability.

Solution

(a) Compartments:
When 5 predators kill 100 prey at once, it's natural to track **schools of prey** N_s (each school contains 100 prey) and **pods of predators** P_p (each pod contains 5 predators). One predation event: 1 pod $\times$ 1 school $\rightarrow$ school eliminated, pod gets energy.

(b) Equations:

$$\frac{dN_s}{dt} = r_s N_s - \bar{\beta}_s N_s P_p$$
$$\frac{dP_p}{dt} = \varepsilon \bar{\beta}_s N_s P_p - Z_p P_p$$

Identical structure to standard LV, just in units of schools and pods.

(c) Parameter interpretation:

- r_s: school growth rate (new fish schools form). Same units as r (time^{-1}).
- $\bar{\beta}_s$: consumption rate per pod on schools. One pod kills one school at rate $\bar{\beta}_s$. In absolute fish numbers, $\bar{\beta}_s$ corresponds to eliminating 100 fish/encounter — much larger per-individual impact than standard LV.
- ε: energy conversion efficiency (same as LV, dimensionless).
- Z_p: pod mortality rate (same units as Z).

Key differences in magnitudes: Since each predation event now kills 100 prey (instead of 1), the effective consumption rate per individual predator is $\bar{\beta}_s/5$ (divided by pod size). However, in terms of the school-level model, $\bar{\beta}_s$ is comparable to standard $\bar{\beta}$.

(d) Stability:
Step 1. Same structure as standard LV:

$$A = \begin{pmatrix} 0 & -\bar{\beta}_s \\ \varepsilon\bar{\beta}_s & 0 \end{pmatrix}$$

Step 2. Tr $= 0$, Det $= \varepsilon\bar{\beta}_s^2 > 0$.
Step 3. Eigenvalues: $\lambda_\pm = \pm i\bar{\beta}_s\sqrt{\varepsilon}$. Pure imaginary $\Rightarrow$ neutral oscillations.
Step 4. Same qualitative stability as standard LV. However, because each predation event wipes out 100 prey, the oscillation amplitude in absolute prey numbers is much larger. The system reaches close to extinction of schools during the troughs. In practice, **more extinction risk** than standard LV.
Compare 4-hunt-1 vs 5-hunt-100:

- Both give Tr $= 0$, Det > 0, neutral oscillations.
- Different magnitudes of $|\lambda|$ and oscillation amplitude.
- Mathematical stability class: same. Practical biological stability: different.

9.5 Problem 9.5: Diagonalized Matrix and Eigenvalues (2019 Q2)

Problem

For a 2×2 matrix A with eigenvalues λ_1, λ_2, the diagonalized version is:

$$A' = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$$

(a) Calculate $\mathrm{Tr}(A')$ and $\mathrm{Det}(A')$. Show these equal $\mathrm{Tr}(A)$ and $\mathrm{Det}(A)$.

(b) Using these two equations, solve for λ_1 and λ_2 in terms of Tr and Det.

(c) For a 3×3 matrix, how many independent equations do you need? What are they?

Solution

(a) Tr and Det of A':
Step 1. Trace is the sum of diagonal entries:

$$\mathrm{Tr}(A') = \lambda_1 + \lambda_2$$

Step 2. Determinant (diagonal matrix): product of diagonal entries:

$$\mathrm{Det}(A') = \lambda_1 \cdot \lambda_2 - 0 \cdot 0 = \lambda_1\lambda_2$$

Step 3. The trace and determinant are invariants under similarity transformations (diagonalization is a similarity transform $A' = P^{-1}AP$). Therefore:

$$\mathrm{Tr}(A) = \mathrm{Tr}(A') = \lambda_1 + \lambda_2$$

$$\mathrm{Det}(A) = \mathrm{Det}(A') = \lambda_1\lambda_2$$

(b) Solve for λ_1, λ_2:
Step 1. We have two equations:

$$\lambda_1 + \lambda_2 = T \quad (T = \mathrm{Tr})$$

$$\lambda_1\lambda_2 = D \quad (D = \mathrm{Det})$$

Step 2. These are Vieta's formulas for the roots of $\lambda^2 - T\lambda + D = 0$.
Step 3. Apply quadratic formula:

$$\lambda = \frac{T \pm \sqrt{T^2 - 4D}}{2}$$

This recovers the standard eigenvalue formula from two algebraic invariants.

(c) 3×3 matrix:
Step 1. For 3×3 with eigenvalues $\lambda_1, \lambda_2, \lambda_3$, the characteristic polynomial is degree 3:

$$\lambda^3 - (\lambda_1 + \lambda_2 + \lambda_3)\lambda^2 + (\lambda_1\lambda_2 + \lambda_1\lambda_3 + \lambda_2\lambda_3)\lambda - \lambda_1\lambda_2\lambda_3 = 0$$

Step 2. Three invariants needed:

- $I_1 = \lambda_1 + \lambda_2 + \lambda_3 = \mathrm{Tr}(A)$
- $I_2 = \lambda_1\lambda_2 + \lambda_1\lambda_3 + \lambda_2\lambda_3 =$ sum of 2×2 principal minors
- $I_3 = \lambda_1\lambda_2\lambda_3 = \mathrm{Det}(A)$

Step 3. Trace and Det give I_1 and I_3; I_2 is a new quantity (not just Tr and Det).
Step 4. For $n \times n$: need n invariants (n coefficients of characteristic polynomial). Tr and Det always appear but for $n \geq 3$ you need $n - 2$ additional ones.

10. Quick Reference: Common Tricks & Patterns

Stability Decision Tree

1. Find fixed point: set $d\vec{X}/dt = 0$.
2. If FP has any zero components $\rightarrow$ **must use Jacobian** (not interaction matrix).
3. If interactions are nonlinear (Type II, MM) $\rightarrow$ **must use Jacobian**.
4. Otherwise either method works at non-trivial FPs.
5. Compute eigenvalues from Tr and Det: $\lambda_\pm = \frac{\mathrm{Tr}}{2}\left[1 \pm \sqrt{1 - 4\mathrm{Det}/\mathrm{Tr}^2}\right]$.
6. Sign analysis:
   - $\mathrm{Tr} < 0$, $\mathrm{Det} > 0$, $\mathrm{Tr}^2 > 4\,\mathrm{Det}$: stable node.
   - $\mathrm{Tr} < 0$, $\mathrm{Det} > 0$, $\mathrm{Tr}^2 < 4\,\mathrm{Det}$: stable spiral.
   - $\mathrm{Det} < 0$: saddle (unstable).
   - $\mathrm{Tr} > 0$, $\mathrm{Det} > 0$: unstable.
   - $\mathrm{Tr} = 0$, $\mathrm{Det} > 0$: neutral oscillations.

Common Modifications and Their Effects

Modification	Effect on Tr	Effect on stability
Logistic growth term	Adds $-r/K$ to diagonal	Stabilizing
Cannibalism	Adds $-\bar{\beta}_c$ to diagonal	Stabilizing
Type II response	N/A (changes structure)	Destabilizing
Predator carrying K_P	Adds $-\varepsilon\bar{\beta}N^*/K_P$	Stabilizing
Symbiosis $+\alpha N_1 N_2$	Off-diagonal positive	Variable
Chronic carrier (no recovery)	Adds $+\alpha N^*$	Destabilizing

Tricks for Computing R_0

- For SIR: $R_0 = \beta N/\gamma$.
- For SIR with deaths: $R_0 = \beta N/(\gamma + \mu)$.
- For SI (no recovery): $R_0 \rightarrow \infty$ trivially (always spreads).
- For multi-strain: each strain has its own R_0, computed as (transmission rate $\times$ susceptible pool)/(removal rate).
- Vaccination with fraction f effective: $R_0^{\mathrm{eff}} = R_0(1-f)$ (herd immunity threshold $f^* = 1 - 1/R_0$).

Common Linear Spaces (Euler-Exact)

Equation	Linear-Euler Space
$\dot{N} = rN$ (exponential)	$\ln N$ vs t
$\dot{N} = rN/t$ (power)	$\ln N$ vs $\ln t$
$\dot{N} = r/t$ (inverse-time)	N vs $\ln t$
$\dot{N} = rN^2$ (super-exp)	$1/N$ vs t
$\dot{N} = rN^p$ $(p \neq 1)$	N^{1-p} vs t
$\dot{N} = rN(1 - N/K)$ (logistic)	$\ln(N/(K-N))$ vs t

When in Doubt on Exam

1. Always start by checking conservation. Reduces equations.
2. Always identify which fixed point matters (often disease-free for spread questions).
3. Always state whether to use A matrix or J^* **before** computing.
4. Compute Tr and Det **first**; they often answer the qualitative question.
5. For Taylor series questions: check if derivatives exist. For "perfect space" questions: ask what makes higher derivatives $= 0$.
6. For QSSA: identify the fast variable (intermediate complex) and set its derivative to 0.
7. For new modifications: ask "does this add a diagonal term?" (changes Tr) or "does this break linearity?" (kills A approach).